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Reusable tortuous path filters for sterilization containers

Abstract

A sterilization container for sterilizing medical instruments that includes a body defining an interior volume and a lid connected to the body and configured to seal the interior volume between the lid and the body. The lid includes an exterior filter structure including a first plurality of ridges and a first plurality of troughs formed between the first ridges. The container includes a filter assembly connected to the lid and including an interior filter structure including a second plurality of ridges and a second plurality of troughs formed between the second ridges. The first plurality of ridges of the exterior filter structure are interposed between the second plurality of ridges of the interior filter structure. The exterior and interior filter structures are configured to form longitudinal channels positioned between the lid and filter assembly to define a rectangular tortuous filter path that inhibits ingress and egress from the interior volume.

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Background/Summary

CROSS-REFERENCE (1) This application is a divisional of U.S. Non-Provisional patent application Ser. No. 17/066,628 filed Oct. 9, 2020, which claims priority to and the benefit of U.S. Provisional Patent Application No. 62/913,446, filed Oct. 10, 2019, titled Reusable Tortuous Path Filters for Sterilization Containers, the details of which are hereby incorporated herein by reference.

TECHNICAL FIELD

(1) The present invention generally relates to sterilization containers for maintaining sterility of medical devices stored therein, and more specifically to filters forming a tortuous path that permit ingress and egress of sterilants into the container while preventing contaminants from entering the container.

BACKGROUND

(2) Sterilization containers are generally used to maintain a sterility of medical devices stored therein prior to use in a medical procedure. Sterilization containers may include a filter or valve system that facilitates an ingress and egress of sterilants into the container to promote a sterilization of the devices included therein. The filter or valve system further inhibits microorganisms and airborne particulates from accessing the container to prevent a contamination of the medical devices enclosed therein.

(3) Accordingly, a need exists for a sterilization container including a tortuous path filter that is including varying structural configurations that are configured and operable to permit a flow of air and sterilant therethrough and increase a flow resistance of contaminants to prevent contamination of the medical devices stored in the container.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective cross-sectional view of an exemplary sterilization container including a body, a lid and a filter assembly according to one or more embodiments shown and described herein;

(2) FIG. 2 is a partial cross-sectional view of the filter assembly secured to the lid of the

sterilization container of FIG. 1 according to one or more embodiments shown and described herein;

(3) FIG. 3 is a side cross-sectional view of a tortuous path filter formed between the lid and the filter assembly of the sterilization container of FIG. 1 according to one or more embodiments shown and described herein;

(4) FIG. 4 is a perspective view of an alternative exemplary sterilization container including a lid and a filter assembly according to one or more embodiments shown and described herein;

(5) FIG. 5 is a perspective view of an alternative exemplary sterilization container including a lid and a filter assembly according to one or more embodiments shown and described herein;

(6) FIG. 6A is a perspective view of another exemplary lid for a sterilization container, with the lid in a compact state according to one or more embodiments shown and described herein;

(7) FIG. 6B is a perspective view of the lid of FIG. 6A, with the lid in an expanded state according to one or more embodiments shown and described herein;

(8) FIG. 6C is a cross-sectional view of the lid of FIG. 6A in the compact state and including a tortuous path filter disposed therein according to one or more embodiments shown and described herein;

(9) FIG. 6D is a cross-sectional view of the lid of FIG. 6A in the expanded state and including the tortuous path filter disposed therein according to one or more embodiments shown and described herein;

(10) FIG. 7A is a perspective view of another exemplary lid for a sterilization container, with the lid in a compact state according to one or more embodiments shown and described herein;

(11) FIG. 7B is a perspective view of the lid of FIG. 7A, with the lid in an expanded state according to one or more embodiments shown and described herein;

(12) FIG. 8 is a side cross-sectional view another exemplary sterilization container including a body and lid forming a tortuous path filter along corresponding sidewalls according to one or more embodiments shown and described herein;

(13) FIG. 9A is a side cross-sectional view of the sterilization container of FIG. 8 according to one or more embodiments shown and described herein;

(14) FIG. 9B is a side cross-sectional view of the sterilization container of FIG. 8 according to one or more embodiments shown and described herein;

(15) FIG. 10 is a perspective view of another exemplary sterilization container including a lid and a filter assembly according to one or more embodiments shown and described herein;

(16) FIG. 11 is a cross-sectional view of the sterilization container of FIG. 10 with the lid and filter assembly forming a tortuous path filter according to one or more embodiments shown and described herein;

(17) FIG. 12 is a partial cross-sectional view of the sterilization container of FIG. 10 with the lid and filter assembly including a plurality of dimples according to one or more embodiments shown and described herein;

(18) FIG. 13 is a perspective view of another exemplary sterilization container including a body, a lid, and a filter assembly having a nesting feature according to one or more embodiments shown and described herein;

(19) FIG. 14 is a side view of a pair of the sterilization containers of FIG. 13 stacked atop one another according to one or more embodiments shown and described herein;

(20) FIG. 15 is a perspective view of another exemplary sterilization container including a body, a lid, and a filter assembly having a nesting feature according to one or more embodiments shown and described herein;

(21) FIG. 16 is a side view of a pair of the sterilization containers of FIG. 15 stacked atop one another according to one or more embodiments shown and described herein;

(22) FIG. 17 is an exploded perspective view of another exemplary sterilization container including a body, a lid, and a filter assembly according to one or more embodiments shown and described

herein;

(23) FIG. **18** is an exploded perspective view of the sterilization container of FIG. **17** according to one or more embodiments shown and described herein; and

(24) FIG. **19** is a cross-sectional view of the sterilization container of FIG. **17** with the lid coupled to the filter assembly and forming a tortuous filter path according to one or more embodiments shown and described herein.

DETAILED DESCRIPTION

(25) Reference will now be made in detail to various embodiments of reusable sterilization containers including tortuous path filters for maintaining a sterility of an interior volume of the container, examples of which are illustrated in the accompanying drawings. Whenever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts. Directional terms as used herein—for example upper, lower, inner, outer, top, bottom, side, distal, and proximal—are made only with reference to the figures as drawn and are not intended to imply absolute orientation.

(26) Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

(27) Unless otherwise expressly stated, it is in no way intended that any method set forth herein be construed as requiring that its steps be performed in a specific order, nor that with any apparatus specific orientations be required. Accordingly, where a method claim does not actually recite an order to be followed by its steps, or that any apparatus claim does not actually recite an order or orientation to individual components, or it is not otherwise specifically stated in the claims or description that the steps are to be limited to a specific order, or that a specific order or orientation to components of an apparatus is not recited, it is in no way intended that an order or orientation be inferred, in any respect. This holds for any possible non-express basis for interpretation, including: matters of logic with respect to arrangement of steps, operational flow, order of components, or orientation of components; plain meaning derived from grammatical organization or punctuation, and; the number or type of embodiments described in the specification.

(28) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the invention belongs. The terminology used in the description herein is for describing particular embodiments only and is not intended to be limiting. As used in the specification and appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Thus, for example, reference to “a” component includes aspects having two or more such components, unless the context clearly indicates otherwise.

(29) As used herein, the terms “horizontal,” “vertical,” “distal,” “proximal,” “longitudinal” and “lateral” are relative terms only, are indicative of a general relative orientation only, and do not necessarily indicate perpendicularity. These terms also may be used for convenience to refer to orientations used in the figures, which orientations are used as a matter of convention only and are not intended as characteristic of the devices shown. The terms “interior” and “exterior” are in reference to an interior volume of the sterilization containers, with “interior” facing toward the interior volume and “exterior” facing away from the interior volume. The present invention and the embodiments thereof to be described herein may be used in any desired orientation. Moreover, horizontal and vertical walls need generally only be intersecting walls, and need not be perpendicular.

(30) I. Sterilization Container with Rectangular Tortuous Path Filter

(31) Referring now to FIGS. **1-3**, one embodiment of a sterilization container **100** is depicted

including a rectangular tortuous path filter that is configured and operable to permit ingress and egress of airflow and sterilants therethrough while inhibiting admission of contaminants into the sterilization container **100**. Specifically referring to FIG. **1**, the sterilization container **100** includes a body **102**, a lid **110**, and a filter assembly **130**. The body **102** serves as a base of the sterilization container **100** and includes a bottom wall **104** and a plurality of sidewalls **106** extending from the bottom wall **104**, thereby defining an interior volume **108** therebetween. The plurality of sidewalls **106** of the body **102** extend from the bottom wall **104** to an open, top end **107** such that a longitudinal height of each of the plurality of sidewalls **106** extends between the open, top end **107** and the bottom wall **104** (i.e. in the $\pm$ -Z direction of the coordinate axes in the figures). In the embodiment, the bottom wall **104** and the plurality of sidewalls **106** are integrally formed with one another such that the body **102** is a unitary structure. However, it should be understood that in other embodiments the bottom wall **104** and/or one or more of the plurality of sidewalls **106** may be separate components secured to one another to form the body **102** of the sterilization container **100**. (32) In the present example, the bottom wall **104** of the body **102** is generally rectangular such that the interior volume **108** formed by the bottom wall **104** and the plurality of sidewalls **106** is rectangularly shaped. Accordingly, the interior volume **108** of the body **102** is sized to receive one or more devices therein, such as, for example, medical apparatuses, instruments, and/or the like. It should be understood that in other embodiments the bottom wall **104** and the interior volume **108** formed within the body **102** may include various other shapes and/or sizes than those shown and described herein without departing from the scope of the present disclosure. In the embodiment, the body **102** is exposed at the open, top end **107** such that the plurality of sidewalls **106** extend distally from the bottom wall **104** and terminate at the open, top end **107** of the body **102**. Accordingly, the interior volume **108** of the body **102** is exposed and operable to receive the one or more devices therein.

(33) Still referring to FIG. **1**, the lid **110** of the sterilization container **100** serves as a top of the sterilization container **100** and is sized and shaped in accordance with a size and shape of the body **102**, such that the lid **110** is generally rectangularly shaped. In particular, the lid **110** includes an exterior surface **112** and an interior surface **114** positioned opposite of the exterior surface **112**. The lid **110** further includes an engagement edge **115** extending about a perimeter of the interior surface **114**. The engagement edge **115** of the lid **110** is sized and shaped relative to a size and shape of the plurality of sidewalls **106** of the body **102** such that the lid **110** is configured to be received over the body **102** and the engagement edge **115** is configured to engage the plurality of sidewalls **106** at the open, top end **107**.

(34) In this instance, the engagement edge **115** of the lid **110** abuts against and forms a seal with the open, top end **107** of the body **102** thereby enclosing the interior volume **108** between the interior surface **114** of the lid **110** and the bottom wall **104** of the body **102**. It should be understood that in some embodiments the engagement edge **115** extending along the interior surface **114** of the lid **110** may include a gasket, a seal, an elastic member, and/or various other features configured to form a secured and/or airtight coupling between the lid **110** and the body **102**. It should be understood that the engagement end **115** may comprise various other structural geometries than those shown and described herein without departing from a scope of the present disclosure, including, for example, channels, ribs, and/or the like. Additionally and/or alternatively, in other embodiments the open, top end **107** of the body **102** may include a gasket, seal, elastic member, and/or other feature configured to form a secured and/or airtight coupling with the lid **110**.

(35) Still referring to FIG. **1**, the lid **110** of the sterilization container **100** further includes a recessed inlet **116**, an outlet opening **118**, and an exterior filter structure **120** disposed along the exterior surface **112**. In particular, the recessed inlet **116** is positioned adjacent to a terminal end of the exterior filter structure **120** and forms a depressed surface along the exterior surface **112** of the lid **110**. It should be understood that the depressed surface formed by the recessed inlet **116** is relative to a planar surface of the exterior surface **112**. The outlet opening **118** of the lid **110** is

positioned adjacent to another terminal end of the exterior filter structure **120** opposite of the recessed inlet **116**. Accordingly, a length of the exterior filter structure **120** along the exterior surface **112** extends between the recessed inlet **116** and the outlet opening **118** such that the exterior filter structure **120** is disposed therebetween.

(36) The outlet opening **118** forms an aperture extending between the exterior surface **112** and the interior surface **114** such that the outlet opening **118** extends through the lid **110**. The exterior filter structure **120** extends laterally across a predetermined width of the exterior surface **112** of the lid **110** (i.e. in the $\pm Y$ direction of the coordinate axes in the figures), and further extends longitudinally across a predetermined length of the exterior surface **112** of the lid **110** (i.e. in the $\pm X$ direction of the coordinate axes in the figures). In the embodiment, the exterior filter structure **120** is integral with the lid **110** such that the exterior filter structure **120** is unitarily-formed with the exterior surface **112**. As described in greater detail herein, it should be understood that a relative position of the recessed inlet **116**, the outlet opening **118**, and/or the exterior filter structure **120** of the lid **110** may be located along various other surfaces and/or positions than those shown and described herein without departing from a scope of the present disclosure.

(37) Still referring to FIG. 1, the filter assembly **130** is sized and shaped to be at least partially received within the recessed inlet **116** of the lid **110**. The lid **110** further includes one or more latch elements **111** formed along the exterior surface **112** that are configured to securely fix the filter assembly **130** to the lid **110**. In the present example, the lid **110** includes a pair of latch elements **111** positioned on the exterior surface **112**, however, it should be understood that in other embodiments the lid **110** may include additional and/or fewer latch elements **111** thereon. For example, the pair of latch elements **111** may include a protrusion, a slot, and/or the like formed on the exterior surface **112** that are sized and shaped to engage corresponding latch elements of the filter assembly **130**. It should be understood that in other embodiments the latch elements **111** of the lid **110** may include various other sizes, shapes, and/or configurations than those shown and described herein. Further, in other embodiments the lid **110** and the filter assembly **130** may form a unitary structure such that the latch elements **111** of the lid **110** and the corresponding latch elements of the filter assembly **130** may be omitted entirely.

(38) The filter assembly **130** of the sterilization container **100** is sized and shaped in accordance with a size and shape of the recessed inlet **116** of the lid **110** such that the filter assembly **130** is generally rectangularly shaped. In particular, the filter assembly **130** includes an exterior surface **132** and an interior surface **134** positioned opposite of the exterior surface **132**. The filter assembly **130** of the sterilization container **100** further includes an inlet **136** disposed along the exterior surface **132**, an outlet opening **138**, and an interior filter structure **140** disposed along the interior surface **134**. In particular, the inlet **136** is positioned adjacent to a terminal end of the interior filter structure **140** and forms a depressed surface along the exterior surface **132** of the filter assembly **130**.

(39) Still referring to FIG. 1, it should be understood that the depressed surface formed by the inlet **136** is relative to a planar surface of the exterior surface **132** of the filter assembly **130**. More specifically, the inlet **136** includes an inlet opening **135** extending through the filter assembly **130** between the exterior surface **132** and the interior surface **134**. In the embodiment, a longitudinal length of the inlet opening **135** (i.e. in the $\pm Y$ direction of the coordinate axes in the figures) extends parallel to a width of the filter assembly **130** and forms a rectangularly-shaped opening through the filter assembly **130**. The inlet **136** of the filter assembly **130** further includes one or more ramps **137** positioned adjacent to the inlet opening **135**. In the embodiment, a longitudinal length of the one or more ramps **137** (i.e. in the $\pm Y$ direction of the coordinate axes in the figures) extend parallel to a width of the inlet opening **135** and forms a sloped-surface from the exterior surface **132** of the filter assembly **130** toward the inlet opening **135**. In the embodiment, the interior filter structure **140** is integral with the filter assembly **130** such that the interior filter structure **140** is unitarily-formed with the interior surface **134**.

(40) Referring now to FIG. 2, the outlet opening **138** of the filter assembly **130** is positioned adjacent to another terminal end of the interior filter structure **140** opposite of the inlet **136**. Accordingly, a length of the interior filter structure **140** along the interior surface **134** extends between the inlet **136** and the outlet opening **138** such that the interior filter structure **140** is disposed therebetween. The interior filter structure **140** extends laterally across a predetermined width of the interior surface **134** of the filter assembly **130** (i.e. in the $\pm Y$ direction of the coordinate axes in the figures), and further extends longitudinally across a predetermined length of the interior surface **134** of the filter assembly **130** (i.e. in the $\pm X$ direction of the coordinate axes in the figures). The filter assembly **130** further includes one or more latch elements **131** formed along the interior surface **134** that are configured to securely fix the filter assembly **130** to the lid **110**.

(41) In the present example, the filter assembly **130** includes a pair of latch elements **131** positioned on the interior surface **134**, however, it should be understood that in other embodiments the filter assembly **130** may include additional and/or fewer latch elements **131** thereon. The pair of latch elements **131** may include, for example, a protrusion, a slot and/or the like formed on the interior surface **134** that are sized and shaped to engage the latch elements **111** of the lid **110**. In particular, the latch element **131** of the filter assembly **130**, formed as a protrusion, is sized and shaped to be received within the latch element **111** of the lid **110**, formed as a slot. Further, the latch element **131** of the filter assembly **130**, formed as a slot, is sized and shaped to receive the latch element **111** of the lid **110**, formed as a protrusion. It should be understood that in other embodiments the latch elements **131** of the filter assembly **130** may include various other sizes, shapes, and/or configurations than those shown and described herein without departing from the scope of the present disclosure.

(42) Still referring to FIG. 2, and as briefly described above, the one or more ramps **137** of the filter assembly **130** are configured to form a sloped and/or curved surface between the exterior surface **132** and the inlet opening **135**. In the present example, the filter assembly **130** of the sterilization container **100** includes a pair of ramps **137** disposed about opposing sides of the inlet opening **135**, however, it should be understood that in other embodiments the filter assembly **130** may include additional and/or fewer ramps **137** than those shown and described herein. The pair of ramps **137** extends parallel to a lateral direction of the inlet opening **135** (i.e. in the $\pm Y$ direction of the coordinate axes in the figures) and is configured to direct airflow and/or other materials, substances, and/or the like toward the inlet opening **135** of the filter assembly **130**. It should be understood that in other embodiments the inlet opening **135** and/or the one or more ramps **137** of the inlet **136** may have various other sizes and/or shapes than those shown and described herein.

(43) The exterior filter structure **120** of the lid **110** includes a plurality of ridges **122** and a plurality of troughs **124** formed therein forming an undulating pattern of ridges **122** and troughs **124**. The plurality of ridges **122** and the plurality of troughs **124** may be formed as a single-piece, monolithic part of the lid **110**. For example, the plurality of ridges **122** and the plurality of troughs **124** may be molded along the rest of the lid **110**. Each trough **124** of the plurality of troughs **124** is formed and positioned between a pair of adjacent ridges **122** of the plurality of ridges **122**, such that a size and shape of each of the troughs **124** corresponds to a relative distance between each of a pair of adjacent ridges **122**. Further, a size and shape of each of the troughs **124** of the plurality of troughs **124** corresponds to a relative height of each of a pair of adjacent ridges **122** forming the trough **124**.

(44) In some embodiments, a distance between adjacent ridges **122** of the exterior filter structure **120** may vary relative to one another along the lid **110** thereby forming an irregular pattern, while in other embodiments a distance between adjacent ridges **122** of the exterior filter structure **120** may be substantially similar to one another thereby forming a regularly continuous pattern. As briefly described above, the exterior filter structure **120** is integral with the lid **110** such that the exterior filter structure **120** is unitarily-formed with the exterior surface **112**. Accordingly, the

plurality of ridges **122** and the plurality of troughs **124** of the exterior filter structure **120** are integrally-formed with the exterior surface **112** such that the ridges **122** and the troughs **124** of the exterior filter structure **120** are unitary with the lid **110**.

(45) Still referring to FIG. 2, the interior filter structure **140** of the filter assembly **130** includes a plurality of ridges **142** and a plurality of troughs **144** formed therein forming an undulating pattern of ridges **142** and troughs **144**. The plurality of ridges **142** and the plurality of troughs **144** may be formed as a single-piece, monolithic part of the filter assembly **130**. For example, the plurality of ridges **142** and the plurality of troughs **144** may be molded along the rest of the filter assembly **130**. Each trough **144** of the plurality of troughs **144** is formed and positioned between a pair of adjacent ridges **142** of the plurality of ridges **142**, such that a size and shape of the troughs **144** corresponds to a relative distance between each of the pair of adjacent ridges **142**. Further, a size and shape of each of the troughs **144** of the plurality of troughs **144** corresponds to a relative height of each of a pair of adjacent ridges **142** forming the trough **144**.

(46) In some embodiments, a distance between adjacent ridges **142** of the interior filter structure **140** may vary relative to one another along the filter assembly **130** thereby forming an irregular pattern, while in other embodiments the distance between adjacent ridges **142** of the interior filter structure **140** may be substantially similar to one another thereby forming a regularly continuous pattern. As briefly described above, the interior filter structure **140** is integral with the filter assembly **130** such that the interior filter structure **140** is a unitarily-formed with the interior surface **134**. Accordingly, the plurality of ridges **142** and the plurality of troughs **144** of the interior filter structure **140** are integrally-formed with the interior surface **134** such that the ridges **142** and the troughs **144** of the interior filter structure **140** are unitary with the filter assembly **130**.

(47) It should be understood that the plurality of ridges **122**, **142** of the filter structures **120**, **140** are sized, shaped and configured to form a tortuous filter path between the filter structures **120**, **140**, respectively. More specifically, the tortuous filter path formed by the plurality of ridges **122**, **142** of the filter structures **120**, **140** includes a plurality of structural impediments between the lid **110** and the filter assembly **130** in the form of physical obstructions, barriers, hurdles and/or the like. The structural impediments formed by the plurality of ridges **122**, **142** may vary based on a predetermined size and shape of the plurality of ridges **122**, **142**, such as, for example, a length, width, and position of each of the plurality of ridges **122**, **142**. Additionally, it should be understood that the plurality of troughs **124**, **144** formed between the plurality of ridges **122**, **142** of the filter structures **120**, **140** are sized, shaped and configured to form a rectangular tortuous filter path between the filter structures **120**, **140**, respectively.

(48) More specifically, the tortuous filter path formed by the plurality of troughs **124**, **144** of the filter structures **120**, **140** includes a plurality of structural impediments between the lid **110** and the filter assembly **130** in the form of physical cavities, recesses, depressions and/or the like. The structural impediments formed by the plurality of troughs **124**, **144** may vary based on a predetermined size and shape of the troughs **124**, **144** such as, for example, a depth, width, and position of each of the plurality of troughs **124**, **144**. In the embodiment, the tortuous filter path formed by the plurality of troughs **124**, **144** of the filter structures **120**, **140** is a rectangular tortuous filter path including a plurality of longitudinal structural impediments between the lid **110** and the filter assembly **130**. For example, the plurality of longitudinal structural impediments forming the rectangular tortuous filter path between the filter structures **120**, **140** may include longitudinal and/or elongated cavities, recesses and/or depressions positioned in a parallel arrangement relative to one another.

(49) Still referring to FIG. 2, with the lid **110** secured to the body **102** and the filter assembly **130** positioned over the lid **110**, the interior filter structure **140** is disposed over the exterior filter structure **120** thereby forming the tortuous path filter therebetween. In other words, the exterior filter structure **120** is configured to mesh with the interior filter structure **140** in response to the lid **110** receiving the filter assembly **130** thereon. In particular, the plurality of ridges **122** of the

exterior filter structure **120** extends into and are received within the plurality of troughs **144** of the interior filter structure **140**. Further, the plurality of ridges **142** of the interior filter structure **140** extend into and are received within the plurality of troughs **124** of the exterior filter structure **120**. In this instance, a rectangular tortuous filter path is formed therebetween including a series of longitudinal, elongated, and/or rectangular channels positioned between the lid **110** and the filter assembly **130**.

(50) Additionally, the pair of latch elements **111** of the lid **110** engages the pair of latch elements **131** of the filter assembly **130**, thereby aligning a position of the recessed inlet **116** of the lid **110** with a position of the inlet **136** of the filter assembly **130**, and more particularly the inlet opening **135**. In this instance, the inlet opening **135** and the recessed inlet **116** collectively form an ingress point for the rectangular tortuous filter path formed by the coupling of the filter structures **120**, **140**. The outlet opening **118** of the lid **110** is further aligned with the outlet opening **138** of the filter assembly **130** such that the outlet openings **118**, **138** collectively form an egress point for the rectangular tortuous path filter formed by the joinder of the filter structures **120**, **140**. In the embodiment, a size and shape of the ingress point formed by the inlet opening **135** and the recessed inlet **116** is substantially similar to a size and shape of the egress point formed by the outlet openings **118**, **138**. Accordingly, the sterilization container **100** of the embodiment is configured to form a consistent inlet cross-sectional area and outlet cross-sectional area between the pair of filter structures **120**, **140**. It should be understood that in other embodiments the size and shape of the ingress point at the inlet opening **135** and the recessed inlet **116** may vary relative to the size and shape of the egress point at the outlet openings **118**, **138** without departing from a scope of the present disclosure.

(51) Referring now to FIG. 3, in this embodiment, the plurality of ridges **122** of the exterior filter structure **120** extend at varying heights relative to one another. In some embodiments, the plurality of ridges **122** may extend at alternating lengths and/or heights in a predetermined pattern while, in other embodiments, the relative heights of the plurality of ridges **122** may be arbitrary and/or random. It should be understood that in other embodiments the plurality of ridges **122** of the exterior filter structure **120** may extend at similar lengths relative to one another. In the present example, the plurality of ridges **122** of the exterior filter structure **120** extend at varying lengths in an alternating pattern relative to an adjacent ridge **122** of the exterior filter structure **120**. A length and/or thickness of the plurality of ridges **122** is determinative of a size and shape of the plurality of troughs **124** of the exterior filter structure **120** formed therebetween. However, it should be understood that the plurality of troughs **124** of the exterior filter structure **120** may extend at varying depths relative to one another in embodiments where the plurality of ridges **122** include similar heights and/or lengths along the exterior filter structure **120**. In some embodiments, the heights of the plurality of ridges **122** may be such that they extend into the plurality of troughs **124** of the lid **110**.

(52) The plurality of ridges **142** of the interior filter structure **140** extend at similar heights and lengths relative to one another. In other embodiments, the plurality of ridges **142** may extend at varying heights and lengths in a predetermined pattern, and the relative heights and lengths of the plurality of ridges **142** may be arranged in an arbitrary (i.e. random) pattern. In the present example, the plurality of ridges **142** of the interior filter structure **140** extend at similar heights in a uniform pattern relative to an adjacent ridge **142** of the interior filter structure **140**. A length and/or thickness of the plurality of ridges **142** in the longitudinal direction (i.e. in the $\pm$ X direction of the coordinate axes in the figures) is determinative of a size and shape of the plurality of troughs **144** of the interior filter structure **140** formed therebetween. However, it should be understood that the plurality of troughs **144** of the interior filter structure **140** may extend at varying depths (i.e. in the $\pm$ Z direction of the coordinate axes in the figures) relative to one another in embodiments where the plurality of ridges **142** include similar heights and/or lengths along the interior filter structure **140** (i.e. in the $\pm$ Z direction of the coordinate axes in the figures). In the illustrated embodiment,

the plurality of troughs **144** have lengths (i.e. in the $\pm$ -X direction of the coordinate axes in the figures) such that the plurality of troughs **144** extend across multiple (e.g., two or more) ridges **142** of the plurality of ridges **142**. In some embodiments, the heights of the plurality of ridges **142** (i.e. in the $\pm$ -Z direction of the coordinate axes in the figures) may be such that they extend into the plurality of troughs **144** of the filter assembly **130**.

(53) Still referring to FIG. 3, the recessed inlet **116** of the lid **110** is sized in accordance with the inlet opening **135** of the filter assembly **130**. As briefly described above, the lid **110** is configured to align the recessed inlet **116** with the inlet opening **135** of the filter assembly **130** in response to the engagement of the latch elements **111** of the lid **110** with the corresponding latch elements **131** of the filter assembly **130**. Further, the outlet opening **118** of the lid **110** is positioned is sized in accordance with the outlet opening **138** of the filter assembly **130**. Accordingly, the lid **110** is further configured to align the outlet opening **118** of the lid **110** with the outlet opening **138** of the filter assembly **130** in response to the engagement of the latch elements **111** of the lid **110** with the corresponding latch elements **131** of the filter assembly **130**.

(54) With the filter assembly **130** secured to the lid **110** and the body **102** enclosed therein, an ingress and egress into the interior volume **108** of the body **102** (see FIG. 1) is facilitated by the inlets **116**, **136**, the rectangular tortuous filter path formed between the filter structures **120**, **140**, and the outlet openings **118**, **138**. In other words, access to the interior volume **108** of the body **102** (see FIG. 1) is inhibited (i.e. limited) as the filter structures **120**, **140** form a series of longitudinal, elongated, and/or rectangular channels that are configured to form a rectangular tortuous filter path including a plurality of structural impediments. In particular, the plurality of troughs **144** of the interior filter structure **140** are sized, shaped and configured to receive the plurality of ridges **122** of the exterior filter structure **120**, and the plurality of troughs **124** of the exterior filter structure **120** are sized, shaped and configured to receive the plurality of ridges **142** of the interior filter structure **140**.

(55) Still referring to FIG. 3, with the filter assembly **130** of the sterilization container **100** coupled to the lid **110** and the filter structures **120**, **140** meshed with one another, the rectangular tortuous path filter formed therebetween is configured to facilitate an ingress and egress of certain materials therethrough and into the interior volume **108** (see FIG. 1) while preventing other materials from traversing therethrough. In particular, the rectangular tortuous path filter formed by the filter structures **120**, **140** is configured to allow an ingress and egress of sterilant from the inlet opening **135** and the recessed inlet **116**, and through the outlet openings **118**, **138**, and further configured to inhibit microorganisms and airborne particulate from traversing through the filter structures **120**, **140** and into the outlet openings **118**, **138**. In other words, the rectangular tortuous filter path formed by the filter structures **120**, **140** is configured to decrease flow resistance of air and sterilant while increasing flow resistance of contaminants into the interior volume **108** of the body **102**.

(56) In other embodiments, the exterior filter structure **120** and/or the interior filter structure **140** may be preassembled with one another prior to coupling the filter assembly **130** to the lid **110** without departing from a scope of the present disclosure. In this instance, the filter assembly **130** includes the pair of filter structures **120**, **140** assembled to one another, with the plurality of ridges **122** and the plurality of troughs **124** meshed with the corresponding plurality of ridges **142** and the plurality of troughs **144**, respectively, prior to coupling the filter assembly **130** with the lid **110**. Accordingly, the filter assembly **130** is configured to be selectively coupled to the lid **110**, and more specifically the connected pair of filter structures **120**, **140** is configured to be assembled onto the lid **110**. In the embodiment, the exterior filter structure **120** is not integrally-formed with the lid **110**. In other words, the exterior filter structure **120** and the interior filter structure **140** are configured to engage the lid **110**, such that the pair of filter structures **120**, **140** is removably secured to at least one of the exterior surface **112** and/or the interior surface **114**.

(57) It should be understood that in other embodiments one or more filter structures may be integrally formed and/or removably assembled onto the bottom wall **104** of the body **102** without

departing from a scope of the present disclosure. In this instance, the sterilization container **100** may include the exterior filter structure **120**, the interior filter structure **140**, and an additional one or more filter structures along the bottom wall **104** of the body **102**. Accordingly, the pair of filter structures **120**, **140** may be configured to facilitate an ingress and egress into the interior volume **108** of the body **102** and the additional one or more filter structures along the bottom wall **104** of the body **102** may be further configured to facilitate an ingress and egress into the interior volume **108**. Alternatively, as described in greater detail herein, in some embodiments a filter structure may be positioned along the interior surface **114** of the lid **110** and/or an exterior surface **132** of the filter assembly **130**.

(58) II. Alternative Sterilization Container with Rectangular Tortuous Path Filter

(59) FIG. **4** schematically depicts an alternative embodiment of a sterilization container **200** that is substantially similar to the sterilization container **100** shown and described above. Except as otherwise described below, the sterilization container **200** may be configured and operable like the sterilization container **100**. For instance, the sterilization container **200** includes a body **202** formed by a bottom wall **204** and a plurality of sidewalls **206** thereby defining an interior volume therein. The sterilization container **200** includes a lid **210** that is secured to the body **202** and includes an exterior surface **212** and an interior surface opposite of the exterior surface **212**. The exterior surface **212** of the lid **210** includes one or more latch elements **211**, a recessed inlet **216**, and a pair of outlet openings **218** disposed thereon. As similarly described in greater detail above with respect to the lid **110** of the sterilization container **100**, the one or more latch elements **211** are sized, shaped and configured to securely couple the lid **210** to one or more other components of the sterilization container **200**, such as a filter assembly **230**. Additionally, the interior volume of the body **202** is generally sized to receive one or more devices therein, such as, for example, medical apparatuses, instruments, and/or the like.

(60) The exterior surface **212** of the lid **210** further includes a pair of filter structures **220** disposed about the recessed inlet **216**, wherein each of the filter structures **220** extends along the exterior surface **212** and is positioned between the recessed inlet **216** and at least one of the pair of outlet openings **218**. In particular, the recessed inlet **216** is positioned adjacent to terminal ends of the pair of filter structures **220** and forms a depressed surface along the exterior surface **212** of the lid **210**. It should be understood that the depressed surface formed by the recessed inlet **216** is relative to a planar surface of the exterior surface **212**. Each of the pair of outlet openings **218** of the lid **210** is positioned adjacent to a terminal end of at least one of the pair of filter structures **220** opposite of the recessed inlet **216**. It should be understood that the pair of filter structures **220** of the lid **210** include a plurality of ridges and a plurality of troughs as similarly described and shown above with respect to the exterior filter structure **120** of the lid **110**.

(61) Still referring to FIG. **4**, the sterilization container **200** further includes a filter assembly **230** having an exterior surface **232** and an interior surface **234** positioned opposite of the exterior surface **232**. The exterior surface **232** of the filter assembly **230** includes an inlet **236** disposed thereon, and in particular the inlet **236** includes an inlet opening **235** and one or more ramps **237** disposed about the inlet opening **235**. In the embodiment, a longitudinal length of the inlet opening **235** of the inlet **236** extends parallel to a width of the filter assembly **230** (i.e. in the $\pm$ X direction of the coordinate axes in the figures) and forms a rectangularly-shaped opening through the filter assembly **230**. In the present example, the inlet **236** of the filter assembly **230** includes a pair of ramps **237** positioned along opposing sides of the inlet opening **235**. The pair of ramps **237** of the filter assembly **230** is configured to form a curved surface between the exterior surface **232** and the inlet opening **235**.

(62) It should be understood that in other embodiments the filter assembly **230** may include additional and/or fewer ramps **237** than those shown and described herein without departing from the scope of the present disclosure. The pair of ramps **237** extends parallel to a length of the inlet opening **235** (i.e. in the $\pm$ X direction of the coordinate axes in the figures) and is configured to

direct airflow and/or other materials, substances, and/or the like toward the inlet opening 235. It should be understood that in other embodiments the inlet opening 235 and/or the one or more ramps 237 of the inlet 236 may have various other sizes and/or shapes than those shown and described herein.

(63) Although not shown, it should be understood that the filter assembly 230 of the sterilization container 200 further includes a pair of filter structures, a pair of outlet openings, and one or more latch elements disposed on the interior surface 234 of the filter assembly 230. Except as otherwise described below, it should be understood that the filter structures, outlet openings, and latch elements of the filter assembly 230 are substantially similar and configured like the interior filter structure 140, the outlet opening 138, and the latch element 131 of the filter assembly 130 shown and described above, respectively. In particular, the pair of outlet openings are positioned along the interior surface 234 at terminal ends of each of the pair of filter structures of the filter assembly 230 such that the outlet openings of the filter assembly 230 are aligned with the pair of outlet openings 218 of the lid 210 when the filter assembly 230 is received over and coupled to the lid 210 of the sterilization container 200. It should be understood that the pair of filter structures of the filter assembly 230 include a plurality of ridges and a plurality of troughs as similarly described and shown above with respect to the interior filter structure 140 of the filter assembly 130. The plurality of ridges and the plurality of troughs in each of the filter structures of the filter assembly 230 are configured to mesh with the plurality of ridges and the plurality of troughs of the pair of filter structures 220 of the lid 210.

(64) Still referring to FIG. 4, securing the interior surface 234 of the filter assembly 230 onto the exterior surface 212 of the lid 210 via the engagement of the corresponding latch elements 211 of the lid 210 and the latch element of the filter assembly 230, respectively, provides a coupling of the filter structures 220 of the lid 210 with the filter structures of the filter assembly 230. In this instance, a pair of tortuous path filters is formed between the lid 210 and the filter assembly 230, and in particular a pair of rectangular tortuous path filters due a rectangularly-shaped profile of the filter structures 220 of the lid 210 and a rectangularly-shaped profile of the filter structures of the filter assembly 230. The sterilization container 200 includes a single ingress point into the tortuous path filters formed by the filter structures 220 of the lid 210 and the filter structures of the filter assembly 230 at the inlet opening 235 and the recessed inlet 216, respectively. Additionally, the sterilization container 200 includes multiple egress points from the tortuous path filters via the pair of outlet openings 218 of the lid 210 and the pair of outlet openings of the filter assembly 230 aligned therewith, respectively.

(65) III. Alternative Sterilization Container with Rectangular Tortuous Path Filter

(66) Referring now to FIG. 5, an alternative embodiment of a sterilization container 300 that is substantially similar to the sterilization container 100, 200 shown and described above is schematically depicted. Except as otherwise described below, the sterilization container 300 may be configured and operable like the sterilization containers 100, 200. For instance, the sterilization container 300 includes a body 302 formed by a bottom wall 304 and a plurality of sidewalls 306 thereby defining an interior volume therein. The sterilization container 300 includes a lid 310 that is secured to the body 302 and includes an exterior surface 312 and an interior surface opposite of the exterior surface 312. The exterior surface 312 of the lid 310 includes one or more latch elements 311, a pair of recessed inlets 316, and an outlet opening 318 disposed thereon. As similarly described in greater detail above with respect to the lid 110 of the sterilization container 100, the one or more latch elements 311 are sized, shaped and configured to securely couple the lid 310 to one or more other components of the sterilization container 300, such as a filter assembly 330. Additionally, the interior volume of the body 302 is generally sized to receive one or more devices therein, such as, for example, medical apparatuses, instruments, and/or the like.

(67) The exterior surface 312 of the lid 310 further includes a pair of filter structures 320 disposed about the outlet opening 318, wherein each of the filter structures 320 extends along the exterior

surface **312** and is positioned between the outlet opening **318** and at least one of the recessed inlets **316**. In particular, the pair of recessed inlets **316** is positioned adjacent to terminal ends of the pair of filter structures **320** and form a depressed surface along the exterior surface **312** of the lid **310**. It should be understood that the depressed surfaces formed by each of the pair of recessed inlets **316** are relative to a planar surface of the exterior surface **312**. The outlet opening **318** of the lid **310** is positioned adjacent to a terminal end of both of the pair of filter structures **320** opposite of the pair of recessed inlets **316**. It should be understood that the pair of filter structures **320** of the lid **310** include a plurality of ridges and a plurality of troughs as similarly described and shown above with respect to the exterior filter structure **120** of the lid **110**.

(68) Still referring to FIG. 5, the sterilization container **300** further includes a filter assembly **330** having an exterior surface **332** and an interior surface **334** positioned opposite of the exterior surface **332**. The exterior surface **332** of the filter assembly **330** includes a pair of inlets **336** disposed thereon corresponding to the pair of recessed inlets **316** of the lid **310**. In particular, the pair of inlets **336** of the filter assembly **330** each includes an inlet opening **335** and one or more ramps **337** disposed about the inlet opening **335**. In the embodiment, a longitudinal length of each of the inlet openings **335** of the pair of inlets **336** extends parallel to a width of the filter assembly **330** (i.e. in the $\pm$ -X direction of the coordinate axes in the figures) and forms a rectangularly-shaped opening through the filter assembly **330**. In the present example, each of the pair of inlets **336** of the filter assembly **330** includes a pair of ramps **337** positioned along opposing sides of the inlet openings **335**, respectively. Each of the pair of ramps **337** of the pair of inlets **336** are configured to form a curved surface between the exterior surface **332** and the corresponding inlet opening **335**.

(69) It should be understood that in other embodiments the filter assembly **330** may include additional and/or fewer ramps **337** than those shown and described herein without departing from the scope of the present disclosure. The pair of ramps **337** extends parallel to a length of the inlet openings **335** and is configured to direct airflow and/or other materials, substances, and/or the like toward the inlet openings **335**. It should be understood that in other embodiments the inlet openings **335** and/or the one or more ramps **337** of the inlets **336** may have various other sizes and/or shapes than those shown and described herein.

(70) Although not shown, it should be understood that the filter assembly **330** of the sterilization container **300** further includes a pair of filter structures, a pair of outlet openings, and one or more latch elements disposed on the interior surface **334** of the filter assembly **330**. Except as otherwise described below, it should be understood that the filter structures, outlet openings, and latch elements of the filter assembly **330** are substantially similar and configured like the interior filter structure **140**, the outlet opening **138**, and the latch element **131** of the filter assembly **130** shown and described above, respectively. In particular, the pair of outlet openings are positioned along the interior surface **334** at terminal ends of each of the pair of filter structures of the filter assembly **330** such that the outlet openings of the filter assembly **330** are aligned with the outlet opening **318** of the lid **310** when the filter assembly **330** is received over and coupled to the lid **310** of the sterilization container **300**. It should be understood that the pair of filter structures of the filter assembly **330** include a plurality of ridges and a plurality of troughs as similarly described and shown above with respect to the interior filter structure **140** of the filter assembly **130**. The plurality of ridges and the plurality of troughs in each of the filter structures of the filter assembly **330** are configured to mesh with the plurality of ridges and the plurality of troughs of the pair of filter structures **320** of the lid **310**.

(71) Still referring to FIG. 5, securing the interior surface **334** of the filter assembly **330** onto the exterior surface **312** of the lid **310** via the engagement of the corresponding latch elements **311** of the lid **310** and the latch element of the filter assembly **330**, respectively, provides a coupling of the filter structures **320** of the lid **310** with the filter structures of the filter assembly **330**. In this instance, a pair of tortuous path filters is formed between the lid **310** and the filter assembly **330**,

and in particular a pair of rectangular tortuous path filters due a rectangularly-shaped profile of the filter structures **320** of the lid **310** and a rectangularly-shaped profile of the filter structures of the filter assembly **330**. The sterilization container **300** includes multiple ingress points into the tortuous path filters formed by the filter structures **320** of the lid **310** and the filter structures of the filter assembly **330** at the inlet openings **335** and recessed inlets **316**, respectively. Additionally, the sterilization container **300** includes a single egress point from the tortuous path filters via the outlet opening **318** of the lid **310** and the pair of outlet openings of the filter assembly **330** aligned therewith, respectively.

(72) IV. Modular Lid with Rectangular Tortuous Path Filter

(73) Referring now to FIGS. **6A-6D**, an alternative embodiment of a modular filter lid assembly **430** that is substantially similar to the filter assembly **130**, **230**, **330** shown and described above is schematically depicted. Except as otherwise described below, the modular filter lid assembly **430** may be configured and operable like the filter assembly **130**, **230**, **330**. Accordingly, it should be understood that the modular filter lid assembly **430** of the present example may be readily incorporated into the sterilization containers **100**, **200**, **300** described above, and more particularly the modular filter lid assembly **430** may be readily utilized with and/or in lieu of the lids **110**, **210**, **310** shown and described above. In many respects, the modular filter lid assembly **430** functions substantially similar to the filter assembly **130**, **230**, **330** described in detail above such that a version of the sterilization containers **100**, **200**, **300** equipped with the modular filter lid assembly **430** of the present example may be configured and operable similar to the sterilization containers **100**, **200**, **300** equipped with the filter assembly **130**, **230**, **330** except for the differences discussed herein.

(74) Referring specifically to FIG. **6A**, the modular filter lid assembly **430** includes a modular body **431** defined by an exterior surface **432**, an interior surface **434**, a pair of sidewalls **435** and a pair of opposing ends **436**. The modular body **431** includes a lateral width extending between the pair of sidewalls **435** (i.e. in the $\pm Y$ direction of the coordinate axes in the figures) and a longitudinal length extending between the pair of opposing ends **436** (i.e. in the $\pm X$ direction of the coordinate axes in the figures). The modular body **431** is formed of a plurality of movable body segments **433** positioned between the pair of opposing ends **436**. Each of the plurality of movable body segments **433** of the modular body **431** is configured to extend outwardly relative to one another to selectively form a plurality of configurations, shapes, profiles, orientations, and/or the like of the modular filter lid assembly **430**. In other words, the plurality of movable body segments **433** are modular such that each of the movable body segments **433** are selectively expandable and/or collapsible relative to one another in the $\pm X$ direction of FIG. **6A**.

(75) As described in greater detail herein, a lateral width of the modular body **431** extending between the pair of sidewalls **435** (i.e. in the $\pm Y$ direction of the coordinate axes in the figures) is fixed and a longitudinal length of the modular body **431** extending between the pair of opposing ends **436** is selectively adjustable (i.e. in the $\pm X$ direction of the coordinate axes in the figures). At least one of the pair of opposing ends **436** of the modular body **431** includes an inlet **437** formed therein. The inlet **437** extends into the modular body **431** of the modular filter lid assembly **430** and is configured to facilitate access to a tortuous path filter of the modular filter lid assembly **430** disposed within the modular body **431**. The modular body **431** of the modular filter lid assembly **430** is depicted in a first, compacted state with the pair of opposing ends **436** separated from one another by a first distance. In this instance, the tortuous path filter formed within the modular body **431** has a first configuration based on the first, compacted state of the modular body **431**.

(76) Referring now to FIG. **6B**, the modular body **431** of the modular filter lid assembly **430** is depicted in a second, expanded state with the pair of opposing ends **436** separated from one another by a second distance that is larger than the first distance shown and described above. In this instance, the tortuous path filter formed within the modular body **431** has a second configuration based on the second, expanded state of the modular body **431**. With the modular body **431** formed

of a plurality of movable body segments **433**, the modular body **431** of the modular filter lid assembly **430** is configured to selectively transition between the first, compacted state and the second, expanded state in response to an outward expansion of each of the plurality of movable body segments **433**. For example, the pair of opposing ends **436** of the modular body **431** may be physically manipulated to manually adjust (i.e. increase and/or decrease) a separation distance positioned therebetween, thereby modifying a longitudinal length of the modular body **431** (i.e. in the $\pm X$ direction of the coordinate axes in the figures). As described in greater detail herein, each of the plurality movable body segments **433** of the modular body **431** are configured to movably expand outwardly and/or collapse inwardly relative one another in response to the physical manipulation of the pair of opposing ends **436** of the modular body **431** in the $\pm X$ direction of FIG. 6A.

(77) By way of illustrative example, in some embodiments the plurality of movable body segments **433** of the modular body **431** are structurally configured to stack atop one another when the modular body **431** is in the first, compacted state, and alternatively unstack relative to one another when the modular body **431** transitions to the second, expanded state. Alternatively, in other embodiments the plurality of movable body segments **433** are configured to be slidably received within an adjacent movable body segment **433** when in the first, compacted state, and alternatively translate externally from an adjacent movable body segment **433** when the modular body **431** transitions to the second, expanded state. In further embodiments, the plurality of movable body segments **433** of the modular body **431** may be individually foldable over itself and/or an adjacent movable body segment **433** when in the first, compacted state, and further unfoldable when the modular body **431** transitions to the second, expanded state. By way of further example, in embodiments the plurality of movable body segments **433** may formed of flexibly depressible materials, such as, for example, polystyrene. In this instance, each of the plurality of movable body segments **433** may be selectively depressed when in the first, compacted state, and alternatively expanded outwardly when the modular body **431** transitions to the second, expanded state.

(78) Accordingly, the plurality of movable body segments **433** are configured to be selectively adjusted to expand and/or collapse the modular body **431** to increase and/or decrease a longitudinal length defined between the pair of opposing ends **436** (i.e. in the $\pm X$ direction of the coordinate axes in the figures). It should be understood that the plurality of movable body segments **433** of the modular body **431** may be configured to transition from the first, compacted state to the second, expanded state in various suitable manners without departing from the scope of the present disclosure. It should further be understood that with a tortuous path filter disposed within the modular body **431**, the tortuous path filter is configured to selectively expand and/or collapse simultaneously with a corresponding expansion and/or collapse of the plurality of movable body segments **433**. In other embodiments, the modular body **431** of the modular filter lid assembly **430** may be configured and operable to selectively adjust the plurality of movable body segments **433** to expand and/or collapse a lateral width of the modular filter lid assembly **430** defined between the pair of opposing sidewalls **435** in the $\pm Y$ direction of FIG. 6B, a thickness defined between the exterior surface **432** and the interior surface **434**, and/or the like.

(79) Referring now to FIGS. 6C-6D, the modular filter lid assembly **430** includes a pair of filter structures **120**, **140** disposed within the modular body **431**, and in particular between the exterior surface **432** and the interior surface **434**. More specifically, the modular filter lid assembly **430** includes the exterior filter structure **120** disposed within the modular body **431** and positioned along a lower interior surface **438** of the modular filter lid assembly **430**. Further, the modular filter lid assembly **430** includes the interior filter structure **140** disposed within the modular body **431** and positioned along the interior surface **134** of the modular filter lid assembly **430**. It should be understood that the filter structures **120**, **140** are substantially similar to the filter structures **120**, **140** shown and described above such that each of the filter structures **120**, **140** of the modular filter lid assembly **430** includes a plurality of ridges **122**, **142** and a plurality of troughs **124**, **144**,

respectively, as similarly described and shown above with respect to the filter structures **120**, **140** of the filter assembly **130**. In this instance, the pair of filter structures **120**, **140**, of the modular filter lid assembly **430** form a tortuous path filter within the modular body **431** that is configured to mesh with one another between the exterior surface **432** and the interior surface **434** of the modular filter lid assembly **430**.

(80) Referring specifically to FIG. **6C**, the modular filter lid assembly **430** is schematically depicted in the first, compacted state such that the pair of filter structures **120**, **140** are positioned in a first configuration with the pair of opposing ends **436** separated from one another by a first distance. In this instance, a longitudinal length of the tortuous path filter formed by the filter structures **120**, **140** is defined between the inlet **437** and the outlet opening **118** with the plurality of ridges **122**, **142** and the plurality of troughs **124**, **144** positioned therebetween. As will be described in greater detail herein, the tortuous path filter formed by the filter structures **120**, **140** is configured to include additional ridges **122**, **142** and troughs **124**, **144** extending between the inlet **437** and the outlet opening **118** when in the second configuration relative to the first configuration shown and described herein.

(81) Referring now to FIG. **6D**, the modular filter lid assembly **430** is schematically depicted in the second, expanded state such that the pair of filter structures **120**, **140** are positioned in a second configuration with the pair of opposing ends **436** separated from one another by a second distance that is greater than the first distance. In this instance, a longitudinal length of the tortuous path filter formed by the filter structures **120**, **140** is defined between the inlet **437** and the outlet opening **118** in the $\pm X$ axis of FIG. **6D** with the plurality of ridges **122**, **142** and the plurality of troughs **124**, **144** positioned therebetween. The tortuous path filter includes additional ridges **122**, **142** and troughs **124**, **144** extending between the inlet **437** and the outlet opening **118** in the second configuration relative to the first configuration in response to actuation of the plurality of movable body segments **433**, and a separation of the pair of opposing ends **436** from one another (i.e. in the $\pm X$ direction of the coordinate axes in the figures). It should be understood that with the inlet **437** positioned along at least one of the pair of opposing ends **436**, a size and/or shape of the inlet **437** remains fixed in the second, expanded state of the modular body **431** relative to the first, compacted state.

(82) In some embodiments, the plurality of ridges **122**, **142** of the filter structures **120**, **140** are structurally configured to be slidably received within one another when in the first configuration. In other embodiments, the plurality of ridges **122**, **142** and the plurality of troughs **124**, **144** of the filter structures **120**, **140** may be individually foldable when in the first configuration. Still in other embodiments the plurality of ridges **122**, **142** may be formed of flexibly depressible materials such that the plurality of ridges **122**, **142** are selectively depressed when in the first configuration. In either instance, the filter structures **120**, **140** are configured to be modularly adjustable to increase and/or decrease a quantity of the plurality of ridges **122**, **142** and the plurality of troughs **124**, **144** included therein in response to an expansion and/or collapse of the modular body **431** as a longitudinal length defined between the pair of opposing ends **436** is adjusted in the $\pm X$ direction of FIGS. **6C-6D**. In other embodiments, the modular filter lid assembly **430** may be configured to selectively expand and/or collapse in various other configurations, shapes, sizes, orientations, and/or the like without departing from the scope of the present disclosure. For example, a lateral width of the modular body **431** defined between the pair of sidewalls **435** may be selectively adjustable (i.e. in the $\pm Y$ direction of the coordinate axes in the figures) in addition to and/or in lieu of a longitudinal length of the modular body **431** defined between the pair of opposing ends **436**.

(83) V. Alternative Modular Lid with Rectangular Tortuous Path Filter

(84) Referring now to FIGS. **7A-7B**, an alternative embodiment of a modular filter lid assembly **530** that is substantially similar to the modular filter lid assembly **430** shown and described above is schematically depicted. Except as otherwise described below, the modular filter lid assembly **530**

may be configured and operable like the modular filter lid assembly **430** such that like reference numerals are used to identify like components. It should be understood that the modular filter lid assembly **530** of the present example may be readily incorporated into the sterilization containers **100, 200, 300** described above, and more particularly the modular filter lid assembly **530** may be readily utilized with the lids **110, 210, 310** shown and described above. In many respects, the modular filter lid assembly **530** functions substantially similar to the modular filter lid assembly **430** except as otherwise described herein.

(85) Referring specifically to FIG. 7A, the modular body **431** of the modular filter lid assembly **530** is defined by the exterior surface **432**, the interior surface **434**, the pair of sidewalls **435** and the pair of opposing ends **436**. The modular body **431** includes a lateral width extending between the pair of sidewalls **435** (i.e. in the $\pm Y$ direction of the coordinate axes in the figures) and a longitudinal length extending between the pair of opposing ends **436** (i.e. in the $\pm X$ direction of the coordinate axes in the figures). Similar to the modular filter lid assembly **430** shown and described above, the modular body **431** of the modular filter lid assembly **530** is formed of the plurality of movable body segments **433** positioned between the pair of opposing ends **436**. Each of the plurality of movable body segments **433** of the modular body **431** is configured to extend outwardly relative to one another to selectively form a plurality of configurations, shapes, profiles, orientations and/or the like of the modular filter lid assembly **530**. In other words, the plurality of movable body segments **433** are modular such that each of the movable body segments **433** are selectively expandable and/or collapsible relative to one another.

(86) At least one of the pair of sidewalls **435** of the modular body **431** includes an inlet **537** formed therein. The inlet **537** extends into the modular body **431** of the modular filter lid assembly **530** and is configured to facilitate access to a tortuous path filter of the modular filter lid assembly **530** disposed within the modular body **431**. The modular body **431** of the modular filter lid assembly **530** is depicted in a first, compacted state with the pair of opposing ends **436** separated from one another by a first distance. In this instance, the tortuous path filter formed within the modular body **431** has a first configuration based on the first, compacted state of the modular body **431**. It should be understood that the modular filter lid assembly **530** may include a tortuous path filter within the modular body **431** similar to the modular filter lid assembly **430** shown and described above (see FIGS. 6C-6D).

(87) Referring now to FIG. 7B, the modular body **431** of the modular filter lid assembly **530** is depicted in a second, expanded state with the pair of opposing ends **436** separated from one another by a second distance that is larger than the first distance shown and described above. In this instance, the tortuous path filter formed within the modular body **431** has a second configuration based on the second, expanded state of the modular body **431**. With the modular body **431** formed of a plurality of movable body segments **433**, the modular body **431** of the modular filter lid assembly **530** is configured to selectively transition between the first, compacted state and the second, expanded state in response to an outward expansion of each of the plurality of movable body segments **433**.

(88) For example, the pair of opposing ends **436** of the modular body **431** may be physically manipulated to manually adjust (i.e. increase and/or decrease) a separation distance therebetween, thereby modifying a longitudinal length of the modular body **431** in the $\pm X$ direction of FIGS. 7A-7B. It should be understood that with the inlet **537** positioned along at least one of the pair of sidewalls ends **535**, a size and/or shape of the inlet **537** is modularly adjustable simultaneously with the modular body **431** of the modular filter lid assembly **530**. In particular, the inlet **537** is configured to expand along the sidewall **435** as the modular body **431** transitions from the first, compacted state to the second, expanded state.

(89) It should be understood that the plurality of movable body segments **433** of the modular body **431** may be configured to transition from the first, compacted state to the second, expanded state in various suitable manners as described above with respect to modular filter lid assembly **430**

without departing from the scope of the present disclosure. It should further be understood that the modular filter lid assembly **530** includes a pair of modular filter structures disposed within the modular body **431** between the exterior surface **432** and the interior surface **434** forming a tortuous path filter therein. The tortuous path filter of the modular filter lid assembly **530** is configured to selectively expand and/or collapse simultaneously with a corresponding expansion and/or collapse of the plurality of movable body segments **433** of the modular body **431**. In other embodiments, the modular filter lid assembly **530** may be configured to selectively expand and/or collapse in various other configurations, shapes, sizes, orientations, and/or the like without departing from the scope of the present disclosure. For example, a lateral width of the modular body **431** defined between the pair of sidewalls **435** may be selectively adjustable (i.e. in the $\pm Y$ direction of the coordinate axes in the figures) in addition to and/or in lieu of a longitudinal length of the modular body **431** defined between the pair of opposing ends **436** (i.e. in the $\pm X$ direction of the coordinate axes in the figures).

(90) VI. Alternative Sterilization Container with Vertical Tortuous Path Filter

(91) FIGS. **8-9** schematically depict an alternative embodiment of a sterilization container **600** that is substantially similar to the sterilization containers **100**, **200**, **300** shown and described above. Except as otherwise described below, the sterilization container **600** may be configured and operable like the sterilization containers **100**, **200**, **300**. For instance, the sterilization container **600** includes a body **602** formed by a bottom wall **604** and a plurality of sidewalls **606** thereby defining an interior volume **608** therein. The sterilization container **600** includes a lid **610** that is secured to the body **602** and includes an exterior surface **612** and an interior surface **614** opposite of the exterior surface **612**. The lid **610** may include a plurality of sidewalls **621** that extend outwardly from the interior surface **614** that are sized and shaped to be received within the interior volume **608** of the body **602**. It should be understood that the lid **610** includes a corresponding quantity of sidewalls **621** relative to the plurality of sidewalls **606** of the body **602**. Additionally, the interior volume **608** of the body **602** is generally sized to receive one or more devices therein, such as, for example, medical apparatuses, instruments, and/or the like.

(92) Referring specifically to FIG. **8**, in some embodiments each of the sidewalls **606** of the body **602** includes a filter structure **601** disposed thereon and positioned along an interior surface **603** of the sidewall **606** that is positioned inwardly into the interior volume **608**. Each of the filter structures **601** of the plurality of sidewalls **606** includes a plurality of ridges **607** extending laterally outward along the interior surface **603** and a plurality of troughs **605** formed along the interior surface **603** and positioned in between a pair of adjacent ridges **607**. It should be understood that the plurality of ridges **607** and the plurality of troughs **605** may be configured and operable similar to the plurality of ridges **122**, **142** and the plurality of troughs **124**, **144** shown and described in greater detail above.

(93) In embodiments, each of the sidewalls **621** of the lid **610** includes a filter structure **620** disposed thereon positioned along an exterior surface **623** of the sidewall **621** that is positioned outwardly. Each of the filter structures **620** of the plurality of sidewalls **621** includes a plurality of ridges **622** extending laterally outward along the exterior surface **623** and a plurality of troughs **624** formed along the exterior surface **623** and positioned in between a pair of adjacent ridges **622**. It should be understood that the plurality of ridges **622** and the plurality of troughs **624** may be configured and operable similar to the plurality of ridges **122**, **142** and the plurality of troughs **124**, **144** shown and described in greater detail above. It should be further understood that in some embodiments a subset of the plurality of sidewalls **621** of the lid **610** includes the filter structure **620** disposed thereon. In this instance, a corresponding subset of the plurality of sidewalls **606** of the body **602** include the filter structure **601** disposed thereon.

(94) Still referring to FIG. **8**, the body **602** of the sterilization container **600** further includes one or more standoffs **609** extending outwardly from the bottom wall **604** and into the interior volume **608** at a predetermined length. As described in greater detail herein, the one or more standoffs **609** of

the body **602** are sized, shaped, and configured to form a gap proximate to the bottom wall **604** to serve as an outlet opening for a tortuous path filter formed in the sterilization container **600** in response to the body **602** coupling the lid **610**. In some embodiments, the body **602** includes a single standoff **609** extending continuously about a perimeter of the bottom wall **604** adjacent to each of the plurality of sidewalls **606**. In other embodiments, the body **602** includes a plurality of standoffs **609** extending about the bottom wall **604** adjacent to each of the plurality of sidewalls **606**. Various other arrangements and/or quantities of the one or more standoffs **609** may be included in the sterilization container **600** without departing from the scope of the present disclosure.

(95) It should be understood that the one or more standoffs **609** of the body **602** are positioned adjacent to each of the sidewalls **606** by a predetermined distance. As described in greater detail herein, the one or more standoffs **609** are positioned along the bottom wall **604** of the body **602** in alignment with a location of the plurality of sidewalls **621** of the lid **610** in response to the lid **610** being received over the body **602**. The lid **610** of the sterilization container **600** further includes one or more standoffs **626** extending outwardly from the each of the plurality of sidewalls **621** at a predetermined length. As will be described in greater detail herein, in some embodiments the lid **610** includes a single standoff **626** extending continuously about an end of the plurality of sidewalls **621** opposite of the interior surface **614**.

(96) Still referring to FIG. **8**, in other embodiments the lid **610** includes a plurality of standoffs **626** extending about an end of each of the plurality of sidewalls **621**. Various other arrangements and/or quantities of the one or more standoffs **626** may be included in the sterilization container **600** without departing from the scope of the present disclosure. As described in greater detail herein, the one or more standoffs **626** are positioned along an end of the plurality of sidewalls **621** in alignment with a location of the one or more standoffs **609** along the bottom wall **604** in response to the lid **610** being received over the body **602**. Although not shown, it should be understood that the body **602** and/or the lid **610** of the sterilization container **600** may further include one or more latch elements for securely fastening the body **602** and the lid **610** to one another.

(97) With the lid **610** disposed over the body **602** of the sterilization container **600**, the plurality of sidewalls **621** are received within the interior volume **608** and positioned adjacent to the plurality of sidewalls **606**. In particular, the exterior surface **623** of the plurality of sidewalls **621** are positioned proximate to and extending toward the interior surface **603** of the plurality of sidewalls **606** when the lid **610** is received in the body **602**. With the filter structure **620** of the plurality of sidewalls **621** positioned along the exterior surface **623** and the filter structure **601** of the plurality of sidewalls **606** positioned along the interior surface **603**, a tortuous path filter is formed between the plurality of sidewalls **606**, **621**. More specifically, the plurality of ridges **607** and the plurality of troughs **605** of the filter structure **601** are configured to mesh with the plurality of ridges **622** and the plurality of troughs **624** thereby forming a tortuous path filter between the lid **610** and the body **602**.

(98) Still referring to FIG. **8**, a plurality of inlets **616** and a plurality of outlet openings **618** are formed between the lid **610** and the body **602** in response to the lid **610** being positioned over the body **602** and the plurality of sidewalls **621** being received within the interior volume **608**. In particular, the plurality of inlets **616** are formed between the interior surface **614**, the plurality of sidewalls **621**, and the plurality of sidewalls **606**. The plurality of outlet openings **618** are formed between the standoff **609** of the bottom wall **604**, the standoff **626** of the plurality of sidewalls **621**, and the plurality of sidewalls **606**. It should be understood that the lid **610** and/or the body **602** is sized, shaped and configured to form the inlet **616** and the outlet opening **618** when securely coupled to one another. In other words, the inlet **616** and the outlet opening **618** comprise a gap formed between the lid **610** and the body **602** in response to the lid **610** and the body **602** securely engaging one another.

(99) It should be understood that the plurality of inlets **616** and/or the plurality of outlet openings

618 may include various sizes, shapes, and/or positions than those shown and described herein without departing from the scope of the present disclosure. Additionally, it should be understood the plurality of inlets **616** and the plurality of outlet openings **618** are located proximate to the plurality of sidewalls **606**, **621** that include the filter structures **601**, **620** disposed thereon. Accordingly, in embodiments with only a subset of the plurality of sidewalls **606**, **621** including the filter structures **601**, **620**, respectively, a gap formed by the inlet **616** and the outlet opening **618** is omitted between the lid **610** and the body **602** along the plurality of sidewalls **606**, **621** not including the filter structures **601**, **620**. In some embodiments, the filter structures **601**, **620** are further configured to form a gasket between the lid **610** and the body **602**.

(100) Referring now to FIG. 9A, the sidewall **621** of the lid **610** is schematically depicted including the one or more standoffs **626** described and shown above. In the embodiment, the sidewall **621** of the lid **610** includes a plurality of standoffs **626** disposed along a bottom end of the sidewall **621**. In particular, each of the plurality of standoff **626** of the sidewall **621** is separated from one another by a gap **627** formed therebetween thereby defining a plurality of recessed standoffs **628** at the gaps **627**. The plurality of recessed standoffs **628** are positioned distally from the standoff **609** of the body **602** along the bottom wall **604** relative to the plurality of standoffs **626**. In this instance, the outlet opening **618** formed between the standoff **609** of the body **602** and the standoff **626** of the sidewall **621** is increased along the plurality of recessed standoffs **628** with the gaps **627** formed therein. It should be understood that in other embodiments the standoff **626** of the sidewall **621** may extend uniformly along a bottom end of the sidewall **621** such that the plurality of recessed standoffs **628** and the plurality of gaps **627** are omitted entirely from the sterilization container **600**.

(101) For example, referring now to FIG. 9B, an alternative sterilization container **600'** is schematically depicted that is substantially similar to the sterilization container **600** shown and described above except for the differences explicitly noted herein. Therefore, identical or substantially similar components are marked with the same reference numerals without further discussion. It should be understood that any components and operabilities of the sterilization container **600'** that are not described explicitly below may be the same as the components and operabilities of the sterilization container **600** described above. In particular, the lid **610** of the sterilization container **600'** includes a plurality of sidewalls **621'** including a standoff **626'** extending along a bottom end of the sidewall **621'**. In this instance, the standoff **626'** forms a uniform, planar end of the sidewall **621'** of the lid **610**.

(102) The body **602** of the sterilization container **600'** includes one or more standoffs **609'** extending outwardly from the bottom wall **604** and toward the plurality of sidewalls **621'** of the lid **610**. In particular, the bottom wall **604** of the body **602** includes a plurality of standoffs **609'** such that each of the plurality of standoff **609'** along the bottom wall **604** is separated from one another by a gap **618'** formed therebetween. In this instance, the outlet opening **618** formed between the plurality of standoffs **609'** of the body **602** and the standoff **626'** of the plurality of sidewalls **621'** is increased along the plurality of gaps **618'** formed therebetween. It should be understood that in other embodiments both the standoff **626**, **626'** of the plurality of sidewalls **621**, **621'** and the standoff **609**, **609'** of the body **602** may include various other shapes, sizes, and/or configurations than those shown and described herein to form varying outlet openings **118** between the lid **610** and the body **602** of the sterilization container **600**, **600'** without departing from the scope of the present disclosure.

(103) VII. Alternative Sterilization Container with Radial Tortuous Path Filter

(104) FIGS. 10-12 schematically depict an alternative embodiment of a sterilization container **700** that is substantially similar to the sterilization containers **100**, **200**, **300** shown and described above. Except as otherwise described below, the sterilization container **700** may be configured and operable like the sterilization containers **100**, **200**, **300**. Referring specifically to FIG. 10, the sterilization container **700** includes a body **702** formed by a bottom wall **204** and a plurality of sidewalls **706** defining an interior volume therein. The sterilization container **700** includes a lid **710**

that is secured to the body **702** and includes an exterior surface **712** and an interior surface opposite of the exterior surface **712**. The exterior surface **712** of the lid **710** includes a recessed inlet **716**, and a plurality of outlet openings **718** disposed thereon. It should be understood that in some embodiments the lid **110** may include one or more latch elements similar to the latch elements **111** of the lid **110** described and shown above that are sized, shaped and configured to securely couple the lid **710** to one or more other components of the sterilization container **700**, such as a filter assembly **730**. Additionally, the interior volume of the body **702** is generally sized to receive one or more devices therein, such as, for example, medical apparatuses, instruments, and/or the like.

(105) The exterior surface **712** of the lid **710** further includes a radial filter structure **720** disposed about the recessed inlet **716**, wherein the filter structure **720** extends along the exterior surface **712** and is positioned between the recessed inlet **716** and the plurality of outlet openings **718**. In the embodiment, the filter structure **720** includes a circular shape, profile, and/or configuration. The recessed inlet **716** forms a depressed surface along the exterior surface **712** of the lid **710** and is disposed within the filter structure **720** and the plurality of outlet openings **718**. It should be understood that the depressed surface formed by the recessed inlet **716** is relative to a planar surface of the exterior surface **712**. The plurality of outlet openings **718** of the lid **710** is positioned in an annular array about the filter structure **720** and the recessed inlet **716**, and more specifically the plurality of outlet openings **718** are positioned adjacent to a terminal boundary of the filter structure **720** opposite of the recessed inlet **716**. In other embodiments, the lid **710** may include filter structures with various other suitable shapes, sizes, and/or configurations, such as, for example, a rectangular filter structure.

(106) Still referring to FIG. **10**, the sterilization container **700** further includes a filter assembly **730** having an exterior surface **732** and an interior surface **734** positioned opposite of the exterior surface **732**. The exterior surface **732** of the filter assembly **730** includes a radial inlet **736** disposed thereon, and in particular the radial inlet **736** includes a radial inlet opening **735** and a radial ramp **737** disposed about the radial inlet opening **735**. In the embodiment, a diameter of the radial inlet opening **735** is sized and shaped in accordance with a size and shape of the recessed inlet **716**, and a size and shape of the radial inlet **736** is sized and shaped in accordance with a size and shape of the radial filter structure **720**. Accordingly, a radially-shaped opening is formed through the filter assembly **730** at the radial inlet **736**. The radial ramp **737** of the filter assembly **730** is configured to form a curved surface between the exterior surface **732** and the radial inlet opening **735**.

(107) It should be understood that in other embodiments the filter assembly **730** may include a larger and/or smaller radial inlet **736** than those shown and described herein without departing from the scope of the present disclosure. The radial ramp **737** is configured to direct airflow and/or other materials, substances, and/or the like toward the radial inlet opening **735**. It should be understood that in other embodiments the radial inlet opening **735** and/or the radial ramp **737** of the radial inlet **736** may have various other sizes and/or shapes than those shown and described herein. Although not shown, it should be understood that in some embodiments the filter assembly **730** of the sterilization container **700** may further include one or more latch elements disposed on the interior surface **734** of the filter assembly **730** for securely coupling the filter assembly **730** to the lid **710**. Except as otherwise described below, it should be understood that the latch elements of the filter assembly **730** are substantially similar and configured like the latch element **131** of the filter assembly **730** shown and described above.

(108) Referring now to FIG. **11**, the sterilization container **700** is schematically depicted with the filter assembly **730** received on and coupled to the lid **710** such that the radial inlet **736** is disposed over the recessed inlet **716**. In the embodiment, the radial filter structure **720** of the lid **710** includes a plurality of ledges **722** and a plurality of dimples **724**, and the filter assembly **730** includes a radial filter structure **740** disposed along the interior surface **734**. In particular, the radial filter structure **740** includes a plurality of ledges **742** and a plurality of dimples **744** positioned along the interior surface **734**. Each ledge **722**, **742** of the plurality of ledges **722**, **742** is formed between a

pair of adjacent dimples **724**, **744** of the plurality of dimples **724**, **744**, respectively, such that a size and shape of each of the ledges **722**, **742** is relative to a distance and/or depth between each of the pair of adjacent dimples **724**, **744** of the respective filter structures **720**, **740**.

(109) In some embodiments, the distance between adjacent ledges **722**, **742** of the filter structure **720**, **740** varies relative to one another along the lid **710** and/or the filter assembly **730**, respectively. In other embodiments, the distance between adjacent ledges **722**, **742** of the filter structure **720**, **740**, respectively, may be substantially similar to one another. Securing the interior surface **734** of the filter assembly **730** onto the exterior surface **712** of the lid **710** via engagement of corresponding latch elements of the lid **710** and the filter assembly **730**, respectively, provides a coupling of the filter structure **720** of the lid **710** with the filter structure **740** of the filter assembly **730**. In this instance, a tortuous filter path formed between the lid **710** and the filter assembly **730**, and in particular a radial tortuous filter path due a radially-shaped profile of the filter structures **720**, **740** of the lid **710** and the filter assembly **730**.

(110) Still referring to FIG. **11**, the sterilization container **700** includes a single ingress point into the radial tortuous filter paths formed by the filter structure **720** of the lid **710** and the filter structure **740** of the filter assembly **730** at the radial inlet opening **735** and the recessed inlet **716**, respectively. Additionally, the sterilization container **700** includes multiple egress points from the radial tortuous path filter via the plurality of outlet openings **718** of the lid **710**. It should be understood that the plurality of ledges **722**, **742** of the filter structures **720**, **740** are sized and shaped to generate a plurality of structural impediments between the lid **710** and the filter assembly **730** in the form of physical obstructions, barriers, hurdles and/or the like along a tortuous filter path formed between the filter structures **720**, **740**, respectively. The plurality of structural impediments formed by the plurality of ledges **722**, **742** may vary based on a predetermined size and shape of the ledges **722**, **742**, such as, for example, a length, width and/or position of each of the plurality of ledges **722**, **742**.

(111) Referring now to FIG. **12**, the plurality of dimples **724**, **744** formed between the plurality of ledges **722**, **742** of the filter structures **720**, **740** are sized and shaped to generate a plurality of structural impediments between the lid **710** and the filter assembly **730** in the form of physical cavities, recesses, depressions, troughs, and/or the like along the radial tortuous filter path formed between the filter structures **720**, **740**, respectively. The plurality of structural impediments formed by the plurality of dimples **724**, **744** may vary based on a predetermined size and shape of the dimples **724**, **744** such as, for example, a depth, diameter and/or position of each of the plurality of dimples **724**, **744**. With the lid **710** secured to the body **702** and the filter assembly **730** positioned over the lid **710**, the filter structure **740** is disposed over the filter structure **720** thereby forming a tortuous filter path therebetween.

(112) In other words, the filter structure **720** is configured to mesh with the filter structure **740** in response to the lid **710** receiving the filter assembly **730** thereon. In particular, the plurality of ledges **722** of the filter structure **720** extend at least partially over the plurality of ledges **742** and/or the plurality of dimples **744** of the filter structure **740**. Further, the plurality of ledges **742** of the filter structure **740** extends at least partially over the plurality of ledges **722** and/or the plurality of dimples **724** of the filter structure **720**. It should be understood that in some embodiments the plurality of ledges **722**, **742** of the filter structures **720**, **740** are at least partially inserted into the plurality of dimples **724**, **744** when the filter structure **740** is disposed over the filter structure **720** such that the plurality of dimples **724**, **744** are sized and shaped to at least partially receive the plurality of ledges **722**, **742** therein. In other embodiments, the plurality of ledges **722**, **742** of the filter structures **720**, **740** are configured to engage one another such that the plurality of ledges **722** at least partially abut against the plurality of ledges **742** of the filter structure **740**.

(113) VIII. Stackable Sterilization Containers with Nesting Feature

(114) FIGS. **13-14** schematically depict an alternative embodiment of a sterilization container **800**. It should be understood that the sterilization container **800** may be substantially similar to the

sterilization containers **100, 200, 300, 600, 700** shown and described above. Except as otherwise described below, the sterilization container **800** may be configured and operable like the sterilization containers **100, 200, 300, 600, 700**. Referring specifically to FIG. **13**, the sterilization container **800** includes a body **802** formed by a bottom wall **804**, one or more sidewalls **806**, and one or more end walls **808**. The walls **804, 806, 808** of the body **802** are generally sized and shaped to form an interior volume that is configured to receive one or more devices therein, such as, for example, medical apparatuses, instruments, and/or the like. The body **802** of the sterilization container **800** further includes one or more engagement surfaces **805** along a juncture between the bottom wall **804** and the one or more end walls **808** thereby forming one or more recessed channels **803** extending therethrough. The one or more engagement surfaces **805** positioned between the bottom wall **804** and the end walls **808** extend at varying angles, orientations, and/or configurations therethrough to form the recessed channel **803** of the body **802**. As described in detail herein, the one or more recessed channels **803** provide nesting features for coupling the body **802** to one or more other components of the sterilization container **800**, such as, for example, a lid **810**.

(115) In the present example, the body **802** of the sterilization container **800** includes a pair of opposing end walls **808** defining a longitudinal length of the body **802** in the $\pm$ -X direction of FIG. **13**, and a pair of opposing sidewalls **806** defining a lateral width of the body **802** in the $\pm$ -Y direction of FIG. **13**. The pair of opposing end walls **808** of the body **802** includes a plurality of engagement surfaces **805** extending from the bottom wall **804** and along each of the end walls **808**, thereby defining at least one recessed channel **803** thereon. In the present example the at least one recessed channel **803** extends along a longitudinal length of the bottom wall **804** of the body **802** between the pair of end walls **808** (i.e. in the $\pm$ -X direction of the coordinate axes in the figures) and has a trapezoidal configuration. As described in greater detail herein, the engagement surfaces **805** of the body **802** are sized and shaped in accordance with a size and shape of a component of the sterilization container **800**, such as, for example, a lid **810**. It should be understood that in other embodiments the body **802** of the sterilization container **800** may include additional and/or fewer engagement surfaces **805** at a juncture between the bottom wall **804** and the end walls **808** than those shown and described herein without departing from the scope of the present disclosure. Additionally, in some embodiments the engagement surfaces **805** of the body **802** may include various other sizes, shapes, arrangements, and/or configurations to thereby form recessed channels with varying profiles.

(116) Still referring to FIG. **13**, the sterilization container **800** further includes a lid **810** defined by an exterior surface **812** and an interior surface opposite of the exterior surface **812**. The lid **810** of the sterilization container **800** further includes at least one coupling mechanism **815** disposed along the exterior surface **812**. In particular, the coupling mechanism **815** of the lid **810** extends outwardly from the exterior surface **812** such that the coupling mechanism **815** forms a protrusion thereon. The coupling mechanism **815** of the lid **810** extends along a longitudinal length of the exterior surface **812** in the $\pm$ -X direction of FIG. **13** and is sized and shaped in accordance with a configuration of the recessed channel **803** of the body **802**. As described in greater detail herein, the coupling mechanism **815** provides a nesting feature for coupling the lid **810** to one or more other components of the sterilization container **800**, such as, for example, the body **802**. In particular, the coupling mechanism **815** on the lid **810** of one sterilization container **800** is configured to engage the recessed channel **803** on the body **802** of another sterilization container **800**. In the present example, the lid **810** of the sterilization container **800** includes one coupling mechanism **815** on the exterior surface **812**, however, it should be understood that in other embodiments the lid **810** may include additional coupling mechanisms **815** positioned on the exterior surface **812** and/or other surfaces of the lid **810**.

(117) The lid **810** of the sterilization container **800** further includes an inlet **816** positioned along the coupling mechanism **815**, and in particular at a terminal end of the coupling mechanism **815**. In some embodiments the lid **810** of the sterilization container **800** includes one or more filter

structures disposed within the coupling mechanism **815** such that a tortuous path filter is formed within the coupling mechanism **815**. In this instance, the inlet **816** of the lid **810** is configured to provide an ingress point into the coupling mechanism **815**, and more specifically toward the tortuous filter path positioned within the coupling mechanism **815**. Although not shown, it should be understood that the one or more filter structures and/or the tortuous path filter disposed within the coupling mechanism **815** of the lid **810** may be configured and operable similar to the filter structures **120**, **140** shown and described above. As described in greater detail herein, the inlet **816** of the lid **810** may be positioned along various other regions of the coupling mechanism **815** and/or the exterior surface **812** without departing from the scope of the present disclosure.

(118) Referring now to FIG. **14**, a pair of sterilization containers **800** are schematically depicted in a stacked configuration with a first sterilization container **800** positioned over and received on a second sterilization container **800**. In particular, the second sterilization containers **800** receives the first sterilization container **800** via a corresponding engagement between the coupling mechanism **815** of the second, lower sterilization container **800** and the recessed channel **803** of the first, upper sterilization container **800**, respectively. With the recessed channel **803** sized and shaped in accordance with a size and shape of the coupling mechanism **815**, the first, upper sterilization container **800** is securely coupled to the second, lower sterilization container **800** in response to the recessed channel **803** receiving the coupling mechanism **815** therein.

(119) In particular, the plurality of engagement surfaces **805** at a juncture between the bottom wall **804** and each of the pair of end walls **808** engage the coupling mechanism **815** of the lid **810** to thereby fasten the first, upper sterilization container **800** to the second, lower sterilization container **800**. The plurality of engagement surfaces **805** forming the recessed channel **803** of the body **802** are configured to inhibit a movement (e.g., lateral, longitudinal, and/or the like) of the coupling mechanism **815** of the lid **810** received therein. It should be understood that the first and second sterilization containers **800** are substantially similar and interchangeable with one another such that the description provided herein of a first and second sterilization container **800** is not intended to be limiting.

(120) Still referring to FIG. **13**, the plurality of engagement surfaces **805** of the body **802** and the coupling mechanism **815** of the lid **810** are cooperatively sized, shaped, and configured to form one or more gaps **801** between the pair of sterilization containers **800** in response to at least one sterilization container **800** being stacked atop at least another sterilization container **800**. In this instance, the one or more gaps **801** extending along a longitudinal length of the sterilization containers **800** (i.e. in the $\pm$ -X direction of the coordinate axes in the figures), and more specifically between the plurality of sidewalls **806** of the body **802** on the first, upper sterilization container **800** and the exterior surface **812** of the lid **810** on the second, lower sterilization container **800**. In this instance, a clearance formed by the one or more gaps **801** facilitates an ingress point to the inlet **816** of the second, lower sterilization container **800** with the first, upper sterilization container **800** disposed thereon.

(121) IX. Alternative Stackable Sterilization Containers with Nesting Feature

(122) FIGS. **15-16** schematically depict an alternative embodiment of a sterilization container **900** that is substantially similar to the sterilization container **800** shown and described above. Except as otherwise described below, the sterilization container **900** may be configured and operable like the sterilization container **800** such that like reference numerals are used to identify like components. For instance, referring specifically to FIG. **15**, the sterilization container **900** includes the body **802** formed by the bottom wall **804**, one or more sidewalls **806**, and one or more end walls **808** as described above. The body **802** of the sterilization container **900** further includes one or more engagement surfaces **805** along a juncture between the bottom wall **804** and the one or more end walls **808** thereby forming one or more recessed channels **803** extending therethrough. The one or more engagement surfaces **805** positioned between the bottom wall **804** and the end walls **808** extend at varying angles, orientations, and/or configurations therethrough to form the recessed

channel **803** of the body **802**.

(123) In the present example, the body **802** of the sterilization container **900** includes a pair of opposing end walls **808** defining a longitudinal length of the body **802** in the $\pm$ -X direction of FIG. **15** and a pair of opposing sidewalls **806** defining a lateral width of the body **802** in the $\pm$ -Y direction of FIG. **15**. The pair of opposing end walls **808** of the body **802** includes a plurality of engagement surfaces **805** extending from the bottom wall **804** and along each of the end walls **808**, thereby defining at least one recessed channel **803** thereon. In the present example, the at least one recessed channel **803** extends along a longitudinal length of the bottom wall **804** of the body **802** between the pair of end walls **808** (i.e. in the $\pm$ -X direction of the coordinate axes in the figures) and has a trapezoidal configuration. Although the body **802** of the present example depicts a single recessed channel **803** formed along the bottom wall **804**, it should be understood that in other embodiments additional recessed channels **803** may be included on the body **802**. As described in greater detail above, the engagement surfaces **805** are sized and shaped in accordance with a size and shape of the lid **810**. It should be understood that in other embodiments the body **802** of the sterilization container **900** may include additional and/or fewer engagement surfaces **805** at a juncture between the bottom wall **804** and the end walls **808** than those shown and described herein. Additionally, in some embodiments the engagement surfaces **805** of the body **802** may include various other sizes, shapes, arrangements, and/or configurations to thereby form recessed channels with varying profiles.

(124) Still referring to FIG. **15**, the body **802** of the sterilization container **900** further includes a plurality of engagement surfaces **905** extending from the bottom wall **804** and along each of the pair of sidewalls **806**, thereby defining at least one recessed channel **907** thereon. In the present example, the at least one recessed channel **907** extends along a lateral width of the bottom wall **804** of the body **802** between the pair of sidewalls **806** in the $\pm$ -Y direction of FIG. **15** and has a trapezoidal configuration. The recessed channel **803** extending through a longitudinal length of the body **802** (i.e. in the $\pm$ -X direction of the coordinate axes in the figures) intersects with and forms a connection with the plurality of recessed channels **907** extending through a lateral width of the body **802** along the bottom wall **804** (i.e. in the $\pm$ -Y direction of the coordinate axes in the figures). Although the body **802** of the present example depicts three recessed channel **907** formed along the bottom wall **804**, it should be understood that in other embodiments additional recessed channels **907** may be included on the body **802**. It should further be understood that in other embodiments the body **802** of the sterilization container **900** may include additional and/or fewer engagement surfaces **905** at a juncture between the bottom wall **804** and the sidewalls **906** than those shown and described herein. Additionally, in some embodiments the engagement surfaces **905** of the body **802** may include various other sizes, shapes, arrangements, and/or configurations to thereby form recessed channels with varying profiles.

(125) The sterilization container **900** further includes the lid **810** defined by the exterior surface **812** and an opposite interior surface opposite with the at least one coupling mechanism **815** disposed on and extending outwardly from the exterior surface **812** to form a protrusion thereon. The coupling mechanism **815** of the lid **810** extends along a longitudinal length of the exterior surface **812** in the $\pm$ -X direction of FIG. **15** and is sized and shaped in accordance with a configuration of the recessed channel **803** of the body **802**. Although the lid **810** of the sterilization container **900** is shown and described herein as including one coupling mechanism **815** in other embodiments the lid **810** may include additional coupling mechanisms **815** along the exterior surface **812** and/or other surfaces of the lid **810**.

(126) Still referring to FIG. **15**, the lid **810** of the sterilization container **900** further includes an inlet **916** positioned along the coupling mechanism **815**, and in particular along a sidewall of the coupling mechanism **815** that extends along a longitudinal length of the coupling mechanism **815**. In this instance, the inlet **916** extends on the coupling mechanism **815** between the opposing end walls **808** of the lid **810**. In some embodiments the lid **810** of the sterilization container **900**

includes one or more filter structures disposed within the coupling mechanism **815** such that a tortuous path filter is formed therein and the inlet **916** is configured to provide an ingress point into the tortuous filter path positioned therein. Although not shown, it should be understood that the one or more filter structures and/or the tortuous path filter disposed within the coupling mechanism **815** of the lid **810** may be configured and operable similar to the filter structures **120**, **140** shown and described above.

(127) Referring now to FIG. **16**, a pair of sterilization containers **900** are schematically depicted in a stacked configuration with a first sterilization container **900** positioned over and received on a second sterilization container **900**. In particular, the second sterilization container **900** receives the first sterilization container **900** via a corresponding engagement between the coupling mechanism **815** of the second, lower sterilization container **900** and the recessed channel **803** of the first, upper sterilization container **900**, respectively. With the recessed channel **803** sized and shaped in accordance with a size and shape of the coupling mechanism **815**, the first, upper sterilization container **900** is securely coupled to the second, lower sterilization container **900** in response to the recessed channel **803** receiving the coupling mechanism **815** therein.

(128) In particular, the plurality of engagement surfaces **805** at a juncture between the bottom wall **804** and each of the pair of end walls **808** engage the coupling mechanism **815** of the lid **810** to thereby fasten the first, upper sterilization container **900** to the second, lower sterilization container **900**. The plurality of engagement surfaces **805** forming the recessed channel **803** of the body **802** are configured to inhibit a movement (e.g., lateral, longitudinal, and/or the like) of the coupling mechanism **815** of the lid **810** received therein. It should be understood that the first and second sterilization containers **900** are substantially similar and interchangeable with one another such that the description provided herein of a first and second sterilization container **900** is not intended to be limiting.

(129) Still referring to FIG. **16**, the plurality of engagement surfaces **805** of the body **802** and the coupling mechanism **815** of the lid **810** are cooperatively sized, shaped, and configured to form one or more gaps **801** between the pair of sterilization containers **900** in response to at least one sterilization container **900** being stacked atop at least another sterilization container **900**. In this instance, the one or more gaps **801** extend along a longitudinal length of the sterilization containers **900** in the $\pm$ X direction of FIG. **16**, and more specifically between the pair of sidewalls **806** of the body **802** on the first, upper sterilization container **900** and the exterior surface **812** of the lid **810** on the second, lower sterilization container **900**. In this instance, a clearance formed by the one or more gaps **801** facilitates an ingress point to the inlet **916** of the second, lower sterilization container **900** with the first, upper sterilization container **900** disposed thereon.

(130) Additionally, the plurality of recessed channels **907** included along the pair of sidewalls **806** of the first, upper sterilization container **900** facilitates a plurality of ingress points to the inlet **916** of the second, lower sterilization container **900** with the first, upper sterilization container **900** disposed thereon. With the inlet **916** included on a sidewall of the coupling mechanism **815** of the second, lower sterilization container **900**, access thereto is provided via the plurality of recessed channels **907** included on the pair of sidewalls **806** of the body **802** of the first, upper sterilization container **900**. In this instance, the plurality of recessed channels **907** of the upper sterilization container **900** are configured to align with the inlet **916** of the lower sterilization container **900** in response to the coupling mechanism **815** of the lower sterilization container **900** receiving and/or engaging the recessed channel **803** of the upper sterilization container **900**.

(131) X. Alternative Sterilization Container with Rectangular Tortuous Path Filter

(132) Referring now to FIGS. **17-19**, an alternative embodiment of a sterilization container **1000** that is substantially similar to the sterilization container **100**, **200**, **300** shown and described above is schematically depicted. Except as otherwise described below, the sterilization container **1000** may be configured and operable like the sterilization containers **100**, **200**, **300**. For instance, the sterilization container **1000** includes a body **1002** formed by a bottom wall **1004**, a plurality of

sidewalls **1006**, and an open, top end **1007** that collectively define an interior volume **1008** therein. The interior volume **1008** of the body **1002** is generally sized to receive one or more devices therein, such as, for example, medical apparatuses, instruments, and/or the like. The sterilization container **1000** includes a lid **1010** that is secured to the body **1002** and includes an exterior surface **1012** and an interior surface **1014** opposite of the exterior surface **1012**. In particular, the interior surface **1014** of the lid **1010** is configured to engage the open, top end **1007** of the body **1002**. The exterior surface **1012** of the lid **1010** includes a filter structure housing **1015** and a recessed inlet **1016** disposed thereon. As described in greater detail herein, the filter structure housing **1015** is sized and shaped to receive one or more components of the sterilization container **1000** therein, such as, for example, a filter assembly **1030**.

(133) The sterilization container **1000** further includes the filter assembly **1030** having an exterior surface **1032** and an interior surface **1034** positioned opposite of the exterior surface **1032**. The exterior surface **1032** of the filter assembly **1030** includes an exterior filter structure **1040** disposed thereon that is sized and shaped in accordance with the filter structure housing **1015** of the lid **1010**, such that the exterior filter structure **1040** is configured to be received within the filter structure housing **1015** when the filter assembly **1030** is coupled to the lid **1010**. Except as otherwise described below, it should be understood that the exterior filter structure **1040** of the filter assembly **1030** is substantially similar and configured like the interior filter structure **140** of the filter assembly **130** shown and described above. For instance, the exterior filter structure **1040** of the filter assembly **1030** includes a plurality of ridges and troughs disposed thereon similar to the plurality of ridges **142** and troughs **144** of the interior filter structure **140**. However, the exterior filter structure **1040** is formed along the exterior surface **1032** of the filter assembly **1030**, rather than the interior surface **1034** as shown and described above with respect to the interior filter structure **140** of the filter assembly **130**.

(134) Still referring to FIG. **17**, the exterior surface **1032** of the filter assembly **1030** further includes one or more latch elements **1031** disposed thereon. The one or more latch elements **1031** of the filter assembly **1030** are sized, shaped and configured to securely couple the filter assembly **1030** to the lid **1010** by engaging one or more corresponding features of the lid **1010**, such as, for example, one or more latch elements **1011** positioned along the interior surface **1014** (see FIG. **18**). It should be understood that the one or more latch elements **1031** of the filter assembly **1030** are configured substantially similar to the one or more latch elements **131** of the filter assembly **130** shown and described above.

(135) Referring now to FIG. **18**, the interior surface **1014** of the lid **1010** is depicted including the one or more latch elements **1011** disposed thereon. As similarly described in greater detail above with respect to the lid **110** of the sterilization container **100**, the one or more latch elements **1011** are sized, shaped and configured to securely couple the lid **1010** to one or the filter assembly **1030** of the sterilization container **1000**. The interior surface **1014** of the lid **1010** further includes an interior filter structure **1020** disposed thereon, wherein the interior filter structure **1020** extends along the interior surface **1014** and is positioned between the recessed inlet **1016** and an outlet opening **1018** of the lid **1010**. Except as otherwise described below, it should be understood that the interior filter structure **1020** of the lid **1010** is substantially similar and configured like the exterior filter structure **120** of the lid **110** shown and described above. For instance, the interior filter structure **1020** of the lid **1010** includes a plurality of ridges and troughs disposed thereon similar to the plurality of ridges **122** and troughs **124** of the exterior filter structure **120**. However, the interior filter structure **1020** is formed along the interior surface **1014** of the lid **1010**, rather than the exterior surface **1012** as shown and described above with respect to the exterior filter structure **120** of the lid **110**.

(136) The filter assembly **1030** of the sterilization container **1000** further includes one or more ramps **1037** positioned along the interior surface **1034**. In particular, the one or more ramps **1037** are disposed about the outlet opening **1038** of the filter assembly **1030**, with a longitudinal length

of the one or more ramps **1037** extending parallel to a width of the outlet opening **1038**. The one or more ramps **1037** forms a sloped-surface from the interior surface **1034** of the filter assembly **1030** toward the outlet opening **1038**. The outlet opening **1038** of the filter assembly **1030** is positioned adjacent to a terminal end of the exterior filter structure **1040**. In the present example, the exterior filter structure **1040** of the filter assembly **1030** is integrally-formed with the exterior surface **1032** such that the exterior filter structure **1040** forms a unitary structure with the filter assembly **1030**. (137) Still referring to FIG. **18**, the interior filter structure **1020** of the lid **1010** is integrally-formed with the interior surface **1014** such that the interior filter structure **1020** forms a unitary structure with the lid **1010**. However, it should be understood that in other embodiments one or more of the filter structures **1020**, **1040** may be removably assembled to the lid **1010** such that the interior filter structure **1020** is not integrally-formed with the lid **1010**. In this instance, the interior filter structure **1020** and the exterior filter structure **1040** may be preassembled to one another and configured to be selectively assembled onto the lid **1010**. It should be understood that in other embodiments one or more filter structures may be integrally formed and/or removably assembled onto the bottom wall **1004** of the body **1002** without departing from a scope of the present disclosure. In this instance, the sterilization container **1000** may include the interior filter structure **1020** along the lid **1010**, the exterior filter structure **1040** along the filter assembly **1030**, and an additional filter structure along the bottom wall **1004** of the body **1002** to facilitate an ingress and egress through the interior volume **1008**.

(138) Referring now to FIG. **19**, the interior surface **1014** of the lid **1010** is coupled to the plurality of sidewalls **1006** of the body **1002** by engaging the open, top end **1007**. The exterior surface **1032** of the filter assembly **1030** is secured to the interior surface **1014** of the lid **1010** via a corresponding engagement of the one or more latch elements **1011**, **1031**. In this instance, the filter assembly **1030** is disposed within the interior volume **1008** of the body **1002** with the exterior filter structure **1040** of the filter assembly **1030** positioned relatively beneath the interior filter structure **1020** of the lid **1010**. More specifically, the exterior filter structure **1040** of the filter assembly **1030** is received within the filter structure housing **1015** of the lid **1010** such that the exterior filter structure **1040** meshes with the interior filter structure **1020**, thereby forming a rectangular tortuous filter path therebetween. The outlet opening **1018** of the lid **1010** is coupled to, and aligned with, the outlet opening **1038** of the filter assembly **1030** when the filter assembly **1030** is received within and coupled to the lid **1010** of the sterilization container **1000**.

(139) Securing the exterior surface **1032** of the filter assembly **1030** onto the interior surface **1014** of the lid **1010** via the engagement of the corresponding latch elements **1011** of the lid **1010** and the latch element **1031** of the filter assembly **1030**, respectively, provides a coupling of the interior filter structure **1020** of the lid **1010** with the exterior filter structure **1040** of the filter assembly **1030**. In this instance, a tortuous path filter is formed between the lid **1010** and the filter assembly **1030** within the interior volume **1008** of the body **1002**. In particular, a rectangular tortuous path filter is formed due a rectangularly-shaped profile of the interior filter structure **1020** of the lid **1010** and a rectangularly-shaped profile of the exterior filter structure **1040** of the filter assembly **1030**. The sterilization container **1000** includes an ingress point into the tortuous path filter formed by the interior filter structure **1020** of the lid **1010** and the exterior filter structure **1040** of the filter assembly **1030** at the recessed inlet **1016**. Additionally, the sterilization container **1000** includes a single egress point from the tortuous path filter via the outlet opening **1018** of the lid **1010** and the outlet opening **1038** of the filter assembly **1030** aligned therewith, respectively.

(140) Further aspects are provided by the subject matter in the following clauses: 1. A sterilization container for sterilizing medical instruments, the sterilization container comprising: a body defining an interior volume; a lid connected to the body and configured to seal the interior volume between the lid and the body, the lid comprising an exterior filter structure integrally formed thereon, the exterior filter structure comprising: a first plurality of ridges; and a first plurality of troughs formed between the first plurality of ridges; a filter assembly connected to the lid and comprising an

interior filter structure integrally formed thereon, the interior filter structure comprising: a second plurality of ridges; and a second plurality of troughs formed between the second plurality of ridges; wherein the first plurality of ridges of the exterior filter structure are interposed between the second plurality of ridges of the interior filter structure; and wherein the exterior filter structure and the interior filter structure are configured to form a series of longitudinal channels positioned between the lid and the filter assembly to define a rectangular tortuous filter path that inhibits ingress and egress from the interior volume. 2. The sterilization container of any preceding clause, wherein at least one trough of the second plurality of troughs has a length that extends at least two ridges of the first plurality of ridges. 3. A sterilization container comprising: a body defining an interior volume; a lid coupled to the body and configured to seal the interior volume therebetween, the lid includes a first filter structure comprising: a first plurality of ridges; and a first plurality of troughs formed between the first plurality of ridges; a filter assembly coupled to the lid and configured to seal the first filter structure therebetween, the filter assembly includes a second filter structure comprising: a second plurality of ridges; and a second plurality of troughs formed between the second plurality of ridges; wherein the first filter structure is configured to mesh with the second filter structure such that the first plurality of ridges is coupled to the second plurality of troughs and the first plurality of troughs is coupled to the second plurality of ridges; and wherein the first filter structure and the second filter structure are configured to form a series of longitudinal channels positioned between the lid and the filter assembly to define a rectangular tortuous filter path that inhibits ingress and egress from the interior volume. 4. The sterilization container of any preceding clause, wherein the lid includes a first inlet opening and a first outlet opening disposed about opposing ends of the first filter structure, and the filter assembly includes a second inlet opening and a second outlet opening disposed about opposing ends of the second filter structure. 5. The sterilization container of any preceding clause, wherein the first inlet opening is in communication with the second inlet opening and the first outlet opening is in communication with the second outlet opening in response to the first filter structure meshing with the second filter structure. 6. The sterilization container of any preceding clause, wherein the first inlet opening is a recessed depression along an exterior surface of the lid, and the second inlet opening is an aperture extending through the filter assembly. 7. The sterilization container of any preceding clause, wherein the filter assembly includes one or more ramps disposed about the second inlet opening that define curved surfaces extending from an exterior surface of the filter assembly toward the second inlet opening. 8. The sterilization container of any preceding clause, wherein the one or more ramps are configured to direct airflow toward the second inlet opening of the filter assembly. 9. The sterilization container of any preceding clause, wherein the lid includes a plurality of first filter structures disposed along an exterior surface of the lid and the filter assembly includes a plurality of second filter structures disposed along an interior surface of the filter assembly. 10. The sterilization container of any preceding clause, wherein the body is sized and shaped to receive one or more medical devices within the interior volume. 11. A modular lid of a sterilization container comprising: a modular body including an inlet and a plurality of movable body segments, wherein the modular body is configured to selectively expand and collapse the plurality of movable body segments relative to one another to thereby selectively adjust a longitudinal length of the modular lid; a first filter structure disposed within the modular body and including a first plurality of ridges and a first plurality of troughs formed between the first plurality of ridges; and a second filter structure disposed within the modular body and including a second plurality of ridges and a second plurality of troughs formed between the second plurality of ridges; wherein the first filter structure is to the second filter structure such that the first plurality of ridges is coupled to the second plurality of troughs and the first plurality of troughs is coupled to the second plurality of ridges thereby forming a series of rectangular channels within the modular body to define a rectangular tortuous filter path that inhibits ingress and egress from the sterilization container; and wherein the first filter structure and the second filter structure are configured to expand and collapse in response

to expansion and collapse of the plurality of movable body segments of the modular body. 12. The modular lid of any preceding clause, wherein the first filter structure is configured to expand the first plurality of ridges and the first plurality of troughs in response to an expansion of the modular body, and further configured to collapse the first plurality of ridges and the first plurality of troughs in response to a collapse of the modular body. 13. The modular lid of any preceding clause wherein the second filter structure is configured to expand the second plurality of ridges and the second plurality of troughs in response to an expansion of the modular body, and further configured to collapse the second plurality of ridges and the second plurality of troughs in response to a collapse of the modular body. 14. The modular lid of any preceding clause, wherein expanding the first plurality of ridges and the first plurality of troughs includes forming additional ridges and troughs within the first filter structure. 15. The modular lid of any preceding clause, wherein expanding the second plurality of ridges and the second plurality of troughs includes forming additional ridges and troughs within the second filter structure. 16. The modular lid of any preceding clause, wherein the inlet is configured to expand in response to an expansion of the modular body, and further configured to collapse in response to a collapse of the modular body. 17. The modular lid of any preceding clause, wherein the inlet is configured to maintain a fixed configuration in response to an expansion and collapse of the modular body. 18. A sterilization container comprising: a body including a bottom wall and a plurality of sidewalls extending outwardly therefrom to define an interior volume disposed therebetween, each of the plurality of sidewalls includes a first filter structure comprising: a first plurality of ridges; and a first plurality of troughs formed between the first plurality of ridges; a lid coupled to the body and configured to seal the interior volume therein, the lid including a top wall and a plurality of sidewalls extending outwardly therefrom, each of the plurality of sidewalls includes a second filter structure comprising: a second plurality of ridges; and a second plurality of troughs formed between the second plurality of ridges; wherein the plurality of sidewalls of the lid are received within the interior volume and positioned against the plurality of sidewalls of the body such that the first filter structures along the first plurality of sidewalls are configured to mesh with the second filter structures along the second plurality of sidewalls of the lid; and wherein the first filter structures and the second filter structures are configured to form a series of rectangular channels positioned between the lid and the body to define a rectangular tortuous filter path that inhibits ingress and egress from the interior volume. 19. The sterilization container of any preceding clause, wherein the first plurality of ridges of the first filter structures positioned along the first plurality of sidewalls are coupled to the second plurality of troughs of the second filter structures positioned along the second plurality of sidewalls. 20. The sterilization container of any preceding clause, wherein the first plurality of troughs of the first filter structures positioned along the first plurality of sidewalls are coupled to the second plurality of ridges of the second filter structures positioned along the second plurality of sidewalls. 21. The sterilization container of any preceding clause, wherein the lid is configured to form a plurality of inlets between the top wall, the plurality of sidewalls of the lid, and the plurality of sidewalls of the body in response to the lid coupling the base. 22. The sterilization container of any preceding clause, wherein the lid is configured to form a plurality of outlet openings between the bottom wall, the plurality of sidewalls of the lid, and the plurality of sidewalls of the body in response to the lid coupling the base. 23. The sterilization container of any preceding clause, wherein the bottom wall of the body includes one or more standoffs extending outwardly therefrom adjacent to the plurality of sidewalls, wherein the one or more standoffs at least partially define the plurality of outlet openings. 24. The sterilization container of any preceding clause, wherein the plurality of sidewalls of the lid includes one or more standoffs extending outwardly therefrom opposite of the top wall, wherein the one or more standoffs at least partially define the plurality of outlet openings. 25. A sterilization container comprising: a body defining an interior volume; a lid coupled to the body and configured to seal the interior volume therebetween, the lid includes a first radial filter structure comprising: a first plurality of dimples; and a first plurality of ledges formed about the

first plurality of dimples; a filter assembly coupled to the lid and configured to seal the first radial filter structure therebetween, the filter assembly includes a second radial filter structure comprising: a second plurality of dimples; and a second plurality of ledges formed about the second plurality of dimples; wherein the first radial filter structure is configured to mesh with the second radial filter structure such that the first plurality of dimples is coupled to the second plurality of dimples and the first plurality of ledges is coupled to the second plurality of ledges; and wherein the first radial filter structure and the second radial filter structure are configured to form a series of recessed depressions positioned between the lid and the filter assembly to define a radial tortuous filter path that inhibits ingress and egress from the interior volume. 26. The sterilization container of any preceding clause, wherein the lid includes a first inlet opening disposed at a center of the first radial filter structure and a plurality of first outlet openings disposed about a boundary of the first radial filter structure. 27. The sterilization container of any preceding clause, wherein the filter assembly includes a second inlet opening disposed at a center of the second radial filter structure and a second outlet opening disposed about a boundary of the second radial filter structure. 28. The sterilization container of any preceding clause, wherein the first inlet opening is in communication with the second inlet opening and the plurality of first outlet openings are in communication with the second outlet opening in response to the first radial filter structure meshing with the second radial filter structure. 29. The sterilization container of any preceding clause, wherein the first inlet opening is a recessed depression along an exterior surface of the lid, and the second inlet opening is an aperture extending through the filter assembly. 30. The sterilization container of any preceding clause, wherein the filter assembly includes one or more ramps disposed about the second inlet opening that define curved surfaces extending from an exterior surface of the filter assembly toward the second inlet opening. 31. The sterilization container of any preceding clause, wherein the one or more ramps are configured to direct airflow toward the second inlet opening of the filter assembly. 32. The sterilization container of any preceding clause, wherein the body is sized and shaped to receive one or more medical devices within the interior volume. 33. A stackable sterilization container comprising: a body including a bottom wall and a plurality of sidewalls extending outwardly therefrom to define an interior volume disposed therebetween, wherein the bottom wall includes one or more recessed channels extending between the plurality of sidewalls; and a lid coupled to the body and configured to seal the interior volume therein, the lid including: a top wall; and an engagement mechanism extending outwardly from the top wall, the engagement mechanism includes an inlet and one or more filter structures disposed therein, the engagement mechanism is sized and shaped relative to the one or more recessed channels such that the engagement mechanism is configured to engage the recessed channel; wherein the one or more filter structures are configured to form a series of rectangular channels within the lid to define a rectangular tortuous filter path that inhibits ingress and egress from the interior volume; wherein the one or more recessed channels are configured to form gaps to access the inlet of the lid when the engagement mechanism is received within at least one of the one or more recessed channels. 34. A sterilization container for sterilizing medical instruments, the sterilization container comprising: a body defining an interior volume; a lid connected to the body and configured to seal the interior volume between the lid and the body, the lid comprising an interior filter structure positioned thereon, the interior filter structure comprising: a first plurality of ridges; and a first plurality of troughs formed between the first plurality of ridges; a filter assembly connected to the lid and comprising an exterior filter structure positioned thereon, the exterior filter structure comprising: a second plurality of ridges; and a second plurality of troughs formed between the second plurality of ridges; wherein the first plurality of ridges of the exterior filter structure are interposed between the second plurality of ridges of the interior filter structure; and wherein the interior filter structure and the exterior filter structure are configured to form a series of longitudinal channels positioned between the lid and the filter assembly to define a rectangular tortuous filter path that inhibits ingress and egress from the interior volume. 35. The sterilization

container of any preceding clause, wherein the interior filter structure is integrally-formed on the lid such that the interior filter structure and the lid are a unitary structure. 36. The sterilization container of any preceding clause, wherein the interior filter structure is releasably coupled to the lid such that the interior filter structure is configured to selectively engage the lid. 37. The sterilization container of any preceding clause, wherein the exterior filter structure is integrally-formed on the filter assembly such that the exterior filter structure and the filter assembly are a unitary structure. 38. The sterilization container of any preceding clause, wherein the exterior filter structure is releasably coupled to the filter assembly such that the exterior filter structure is configured to selectively engage the filter assembly. 39. A sterilization container comprising: a body defining an interior volume; a lid coupled to the body and configured to seal the interior volume therebetween, the lid includes a first filter structure disposed within the interior volume and comprising: a first plurality of ridges; and a first plurality of troughs formed between the first plurality of ridges; a filter assembly coupled to the lid and disposed within the interior volume, the filter assembly is configured to seal the first filter structure therebetween and includes a second filter structure comprising: a second plurality of ridges; and a second plurality of troughs formed between the second plurality of ridges; wherein the first filter structure is configured to mesh with the second filter structure such that the first plurality of ridges is coupled to the second plurality of troughs and the first plurality of troughs is coupled to the second plurality of ridges; and wherein the first filter structure and the second filter structure are configured to form a series of longitudinal channels positioned between the lid and the filter assembly to define a rectangular tortuous filter path that inhibits ingress and egress from the interior volume. 40. A sterilization container comprising: a body defining an interior volume; a lid coupled to the body and configured to seal the interior volume therebetween, a filter assembly releasably coupled to the lid, the filter assembly comprising a first filter structure coupled to a second filter structure, wherein the first filter structure includes a first plurality of ridges and a first plurality of troughs formed between the first plurality of ridges, and the second filter structure includes a second plurality of ridges and a second plurality of troughs formed between the second plurality of ridges; wherein the first plurality of ridges mesh with the second plurality of troughs and the first plurality of troughs mesh with the second plurality of ridges, thereby forming a series of longitudinal channels positioned therebetween and defining a rectangular tortuous filter path that inhibits ingress and egress through the filter assembly. 41. The sterilization container of any preceding clause, wherein the filter assembly is configured to selectively engage the lid such that the rectangular tortuous filter path is removable from the lid. 42. The sterilization container of any preceding clause, wherein the filter assembly is coupled to the lid along an exterior surface of the lid such that the filter assembly is external of the interior volume of the body. 43. The sterilization container of any preceding clause, wherein the filter assembly is coupled to the lid along an interior surface of the lid such that the filter assembly is disposed within the interior volume of the body.

(141) It is noted that the terms “substantially” and “about” may be utilized herein to represent the inherent degree of uncertainty that may be attributed to any quantitative comparison, value, measurement, or other representation. These terms are also utilized herein to represent the degree by which a quantitative representation may vary from a stated reference without resulting in a change in the basic function of the subject matter at issue.

(142) For the purposes of describing and defining the present invention it is noted that the term “substantially” is used herein to represent the inherent degree of uncertainty that may be attributed to any quantitative comparison, value, measurement, or other representation. The term “substantially” is used herein also to represent the degree by which a quantitative representation may vary from a stated reference without resulting in a change in the basic function of the subject matter at issue. As such, it is used to represent the inherent degree of uncertainty that may be attributed to any quantitative comparison, value, measurement, or other representation, referring to an arrangement of elements or features that, while in theory would be expected to exhibit exact

correspondence or behavior, may in practice embody something slightly less than exact. (143) While particular embodiments have been illustrated and described herein, it should be understood that various other changes and modifications may be made without departing from the spirit and scope of the claimed subject matter. Moreover, although various aspects of the claimed subject matter have been described herein, such aspects need not be utilized in combination. It is therefore intended that the appended claims cover all such changes and modifications that are within the scope of the claimed subject matter.

Claims

1. A modular lid of a sterilization container comprising: a modular body including an inlet and a plurality of movable body segments, wherein the modular body is configured to selectively expand and collapse the plurality of movable body segments relative to one another to thereby selectively adjust a longitudinal length of the modular lid; a first filter structure disposed within the modular body and including a first plurality of ridges and a first plurality of troughs formed between the first plurality of ridges; and a second filter structure disposed within the modular body and including a second plurality of ridges and a second plurality of troughs formed between the second plurality of ridges; wherein the first filter structure is to the second filter structure such that the first plurality of ridges is coupled to the second plurality of troughs and the first plurality of troughs is coupled to the second plurality of ridges thereby forming a series of rectangular channels within the modular body to define a rectangular tortuous filter path that inhibits ingress and egress from the sterilization container; and wherein the first filter structure and the second filter structure are configured to expand and collapse in response to expansion and collapse of the plurality of movable body segments of the modular body.
2. The modular lid of claim 1, wherein the first filter structure is configured to expand the first plurality of ridges and the first plurality of troughs in response to an expansion of the modular body, and further configured to collapse the first plurality of ridges and the first plurality of troughs in response to a collapse of the modular body.
3. The modular lid of claim 1, wherein the second filter structure is configured to expand the second plurality of ridges and the second plurality of troughs in response to an expansion of the modular body, and further configured to collapse the second plurality of ridges and the second plurality of troughs in response to a collapse of the modular body.
4. The modular lid of claim 3, wherein expanding the first plurality of ridges and the first plurality of troughs includes forming additional ridges and troughs within the first filter structure.
5. The modular lid of claim 3, wherein expanding the second plurality of ridges and the second plurality of troughs includes forming additional ridges and troughs within the second filter structure.
6. The modular lid of claim 1, wherein the inlet is configured to expand in response to an expansion of the modular body, and further configured to collapse in response to a collapse of the modular body.
7. The modular lid of claim 1, wherein the inlet is configured to maintain a fixed configuration in response to an expansion and collapse of the modular body.
8. The modular lid of claim 1, wherein the inlet is a first inlet opening, the modular lid comprising a first outlet opening and the first inlet opening and the first outlet opening disposed at opposite ends of the first filter structure.
9. The modular lid of claim 1, wherein the filter assembly includes a second inlet opening and a second outlet opening disposed at opposite ends of the second filter structure.
10. The modular lid of claim 9, wherein the first inlet opening is in communication with the second inlet opening and the first outlet opening is in communication with the second outlet opening in response to the first filter structure meshing with the second filter structure.

11. The modular lid of claim 10, wherein the first inlet opening is a recessed depression along an exterior surface of the lid, and the second inlet opening is an aperture extending through the filter assembly.

12. The modular lid of claim 11, wherein the filter assembly includes one or more ramps disposed about the second inlet opening that define curved surfaces extending from an exterior surface of the filter assembly toward the second inlet opening.

13. The modular lid of claim 12, wherein the one or more ramps are configured to direct airflow toward the second inlet opening of the filter assembly.

14. The modular lid of claim 1, wherein the lid includes a plurality of first filter structures disposed along an exterior surface of the modular body and the filter assembly includes a plurality of second filter structures disposed along an interior surface of the filter assembly.
